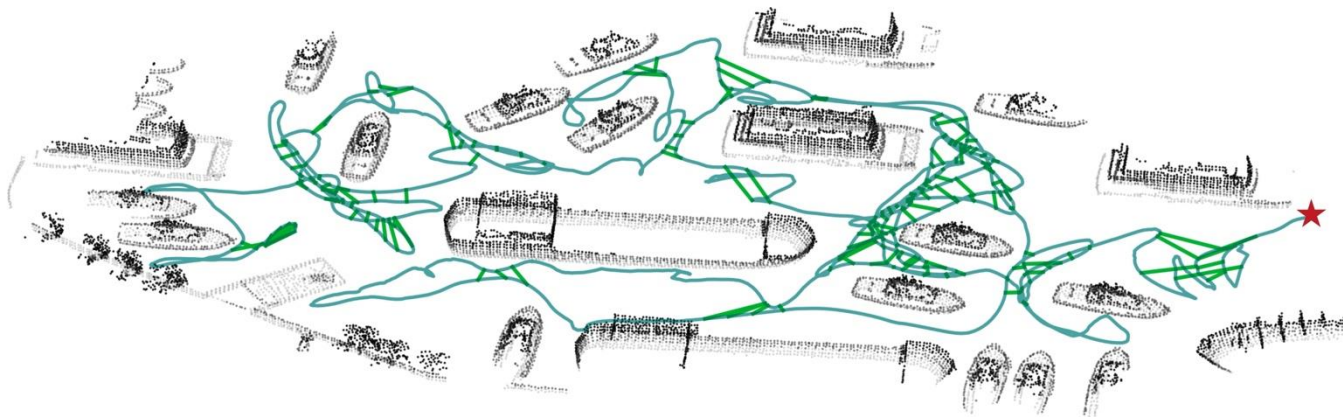
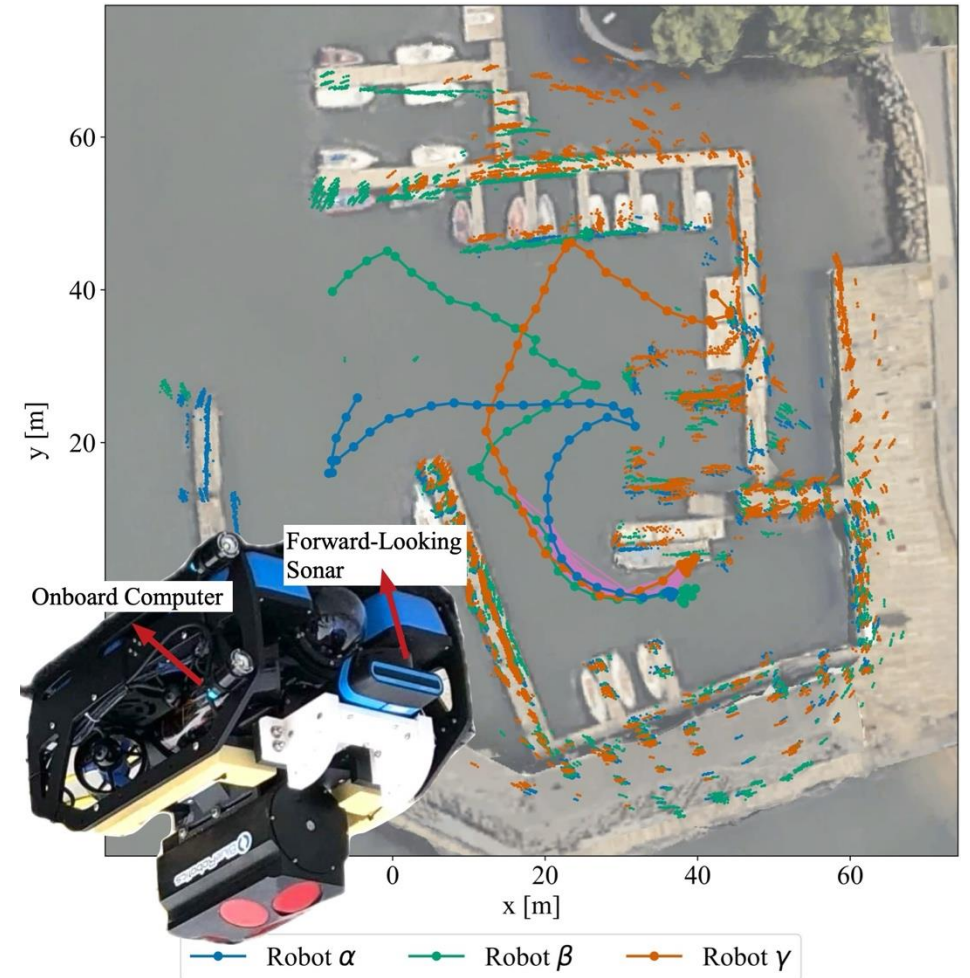


Simultaneous Localization and Mapping (SLAM): From Robot Localization to Underwater Mapping

Yewei Huang
Dartmouth College
AOE 4984 Guest Lecture



— RRT Trajectory	— USV Trajectory	— Loop Closure	• Fixed Landmark	• Virtual Landmark	• Point-cloud
★ Current Pose	→ Goal Position	■ Occupied	□ Free	■ Unknown	



Yewei Huang

Background

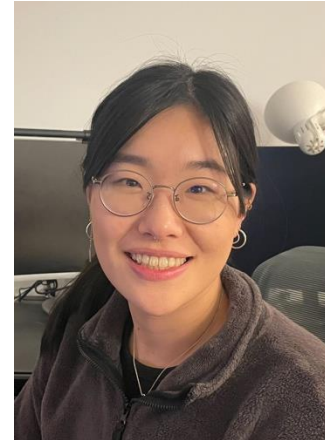
- Postdoctoral researcher in the [Reality and Robotics Lab](#) at the Department of Computer Science, Dartmouth College, advised by [Alberto Quattrini Li](#), Associate Professor.
- Ph.D. in Mechanical Engineering, Stevens Institute of Technology, advised by [Brendan Englot](#), Anson Wood Burchard Endowed Professor and Director of the Stevens Institute for Artificial Intelligence (SIAI).

Research Interest

- **Navigation and perception** of the multi-robot teams, especially in the marine environment.
- **Check my latest work at:**
<https://yewei Huang.github.io/>
- **Check latest work of RFAL at:**
<https://robustfieldautonomylab.github.io/>
- **Check latest work of R-Lab at:**
<https://rlab.cs.dartmouth.edu/home/>



DARTMOUTH



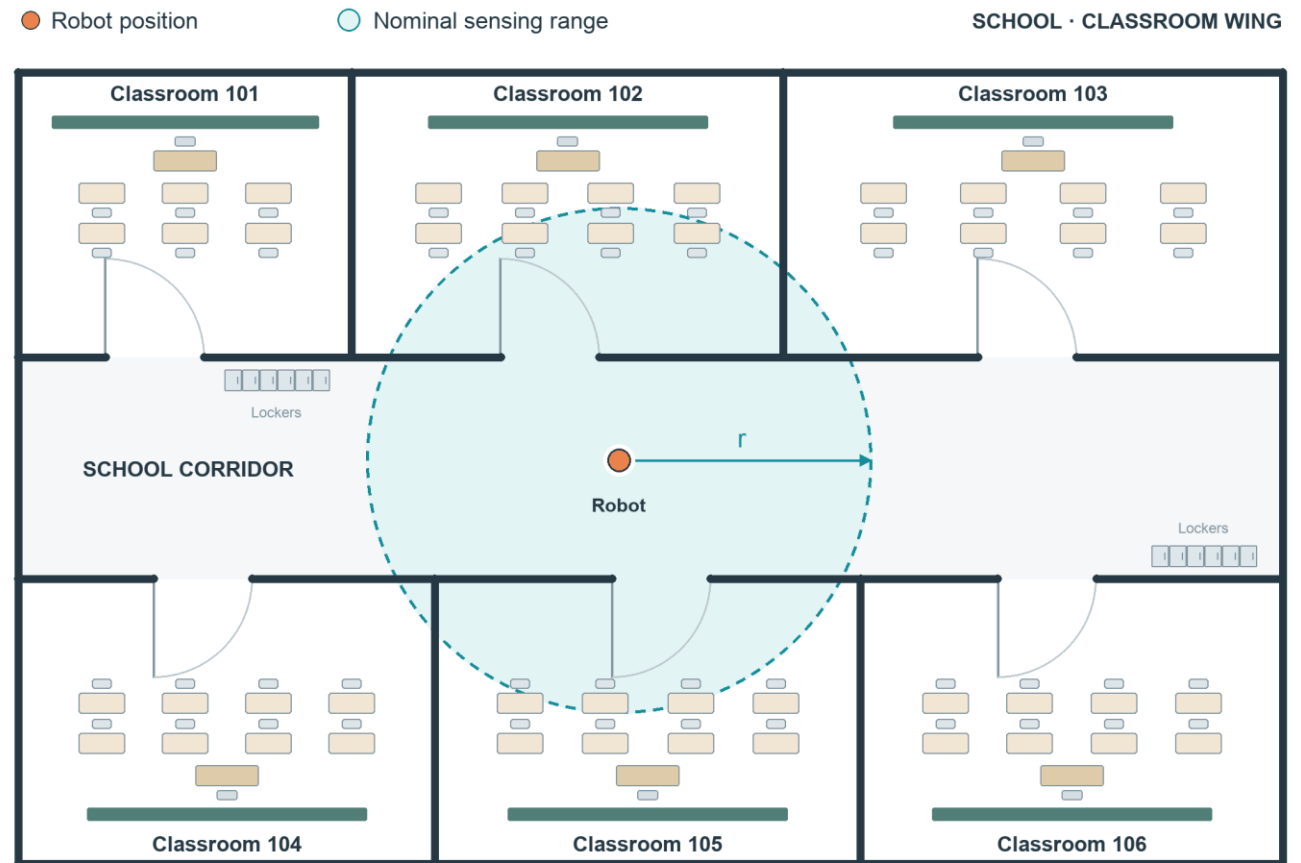
What is Simultaneous Localization and Mapping (SLAM)?

- **SLAM** is the process by which a robot simultaneously builds a map of an **unknown** environment and estimates its own position within that map.

Where am I?
What is around me?



Random robot vacuum
from Internet



What is Simultaneous Localization and Mapping (SLAM)?

How human localize themselves?

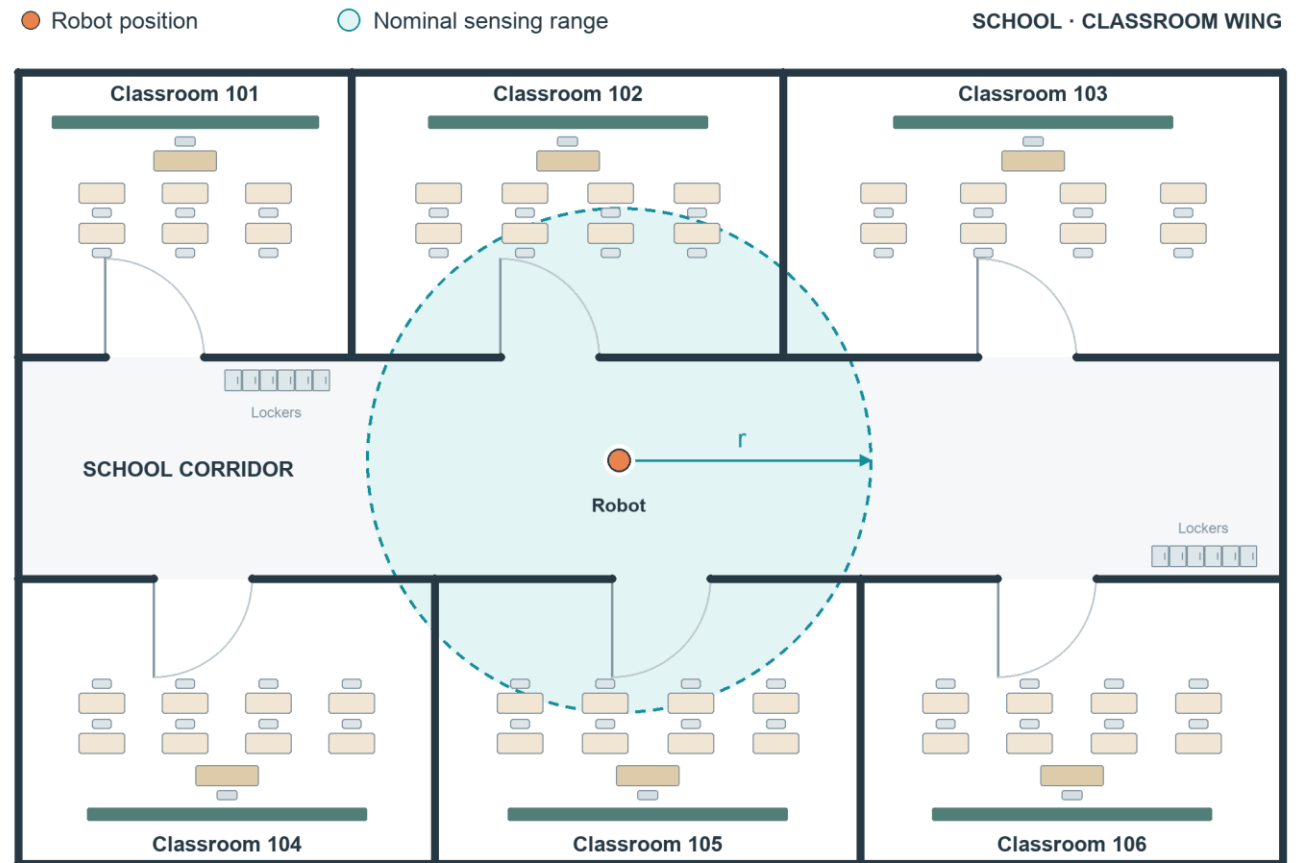
- Look around with eyes and think.

How robot localize themselves?

Where am I?
What is around me?

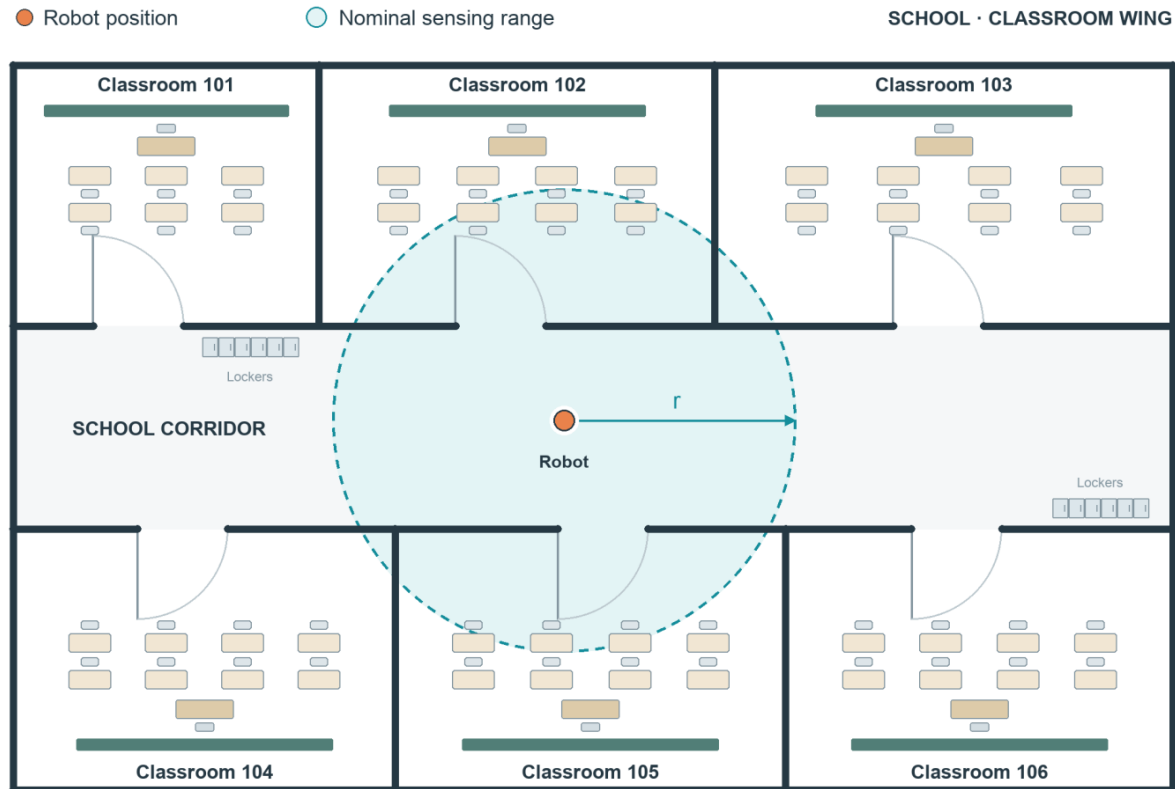


Random robot vacuum
from Internet



How do people find their way?

Think about walking back to this classroom.



Observe

Look around for landmarks.

Recognize

Identify a familiar place.

Connect

Relate that place to your position.

A familiar place gives a reference for your position.

How does a robot find its way?

Sense

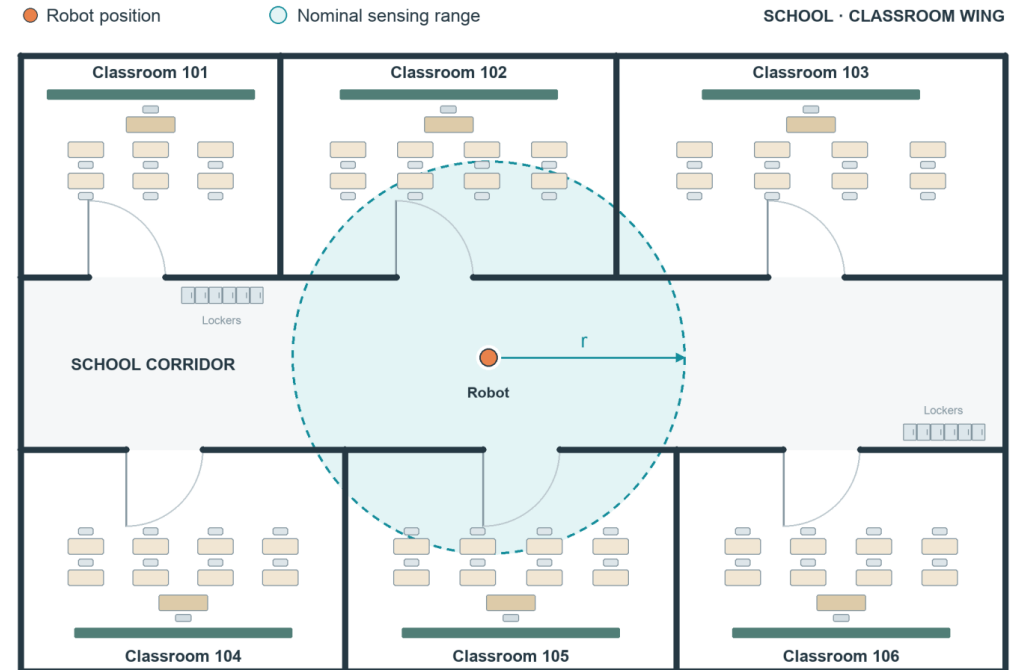
Observe with cameras or sonar.

Estimate

Measure how the robot moved.

Compute

Combine observations onboard.



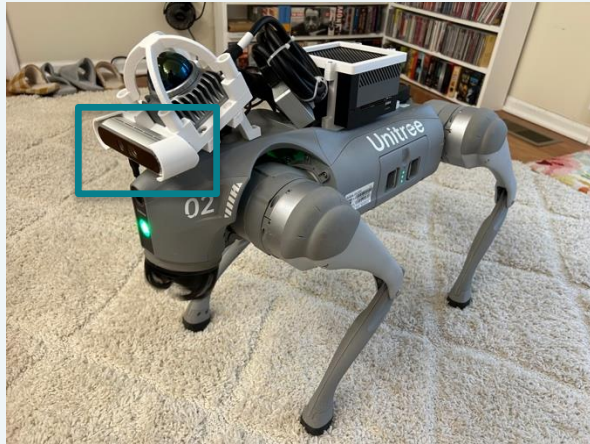
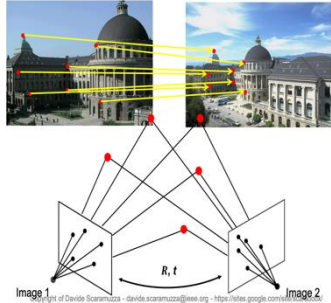
The same two questions: where am I, and what is around me?

Common sensors for SLAM

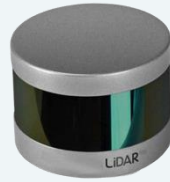
Camera



Intel RealSense



LiDAR



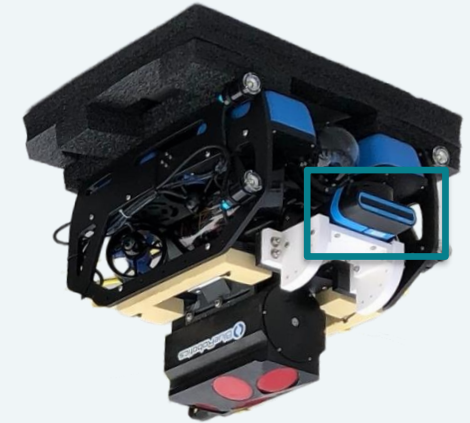
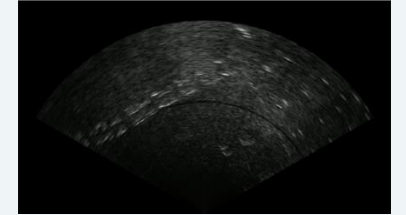
VLP-16



Imaging Sonar



Oculus 750d



IMU

Angular rate + specific force

GNSS / pressure sensor

Global position / depth

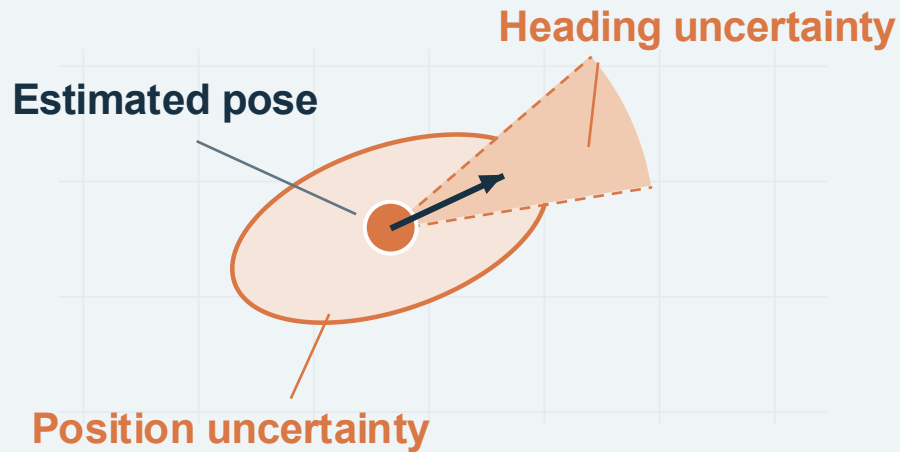
Wheel encoders / DVL

Wheel rotation / velocity*

What is uncertainty in localization and mapping?

Localization uncertainty

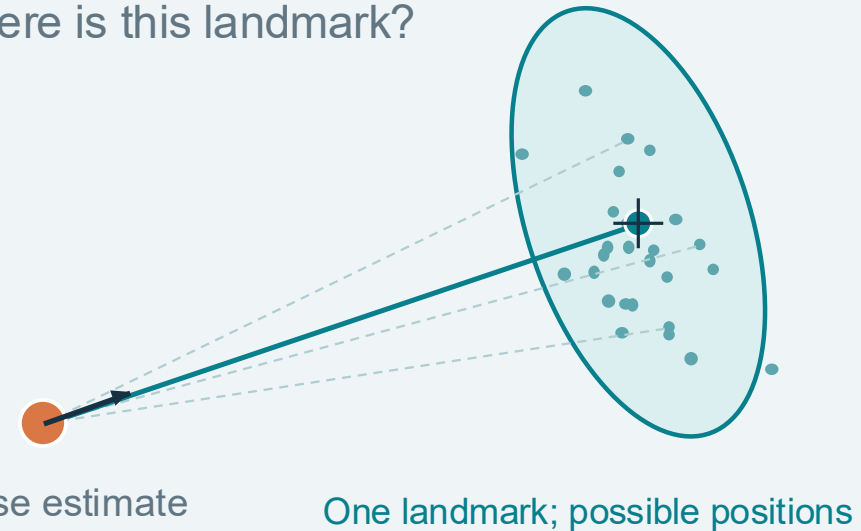
Where am I?



Motion and sensor noise make position and heading uncertain.

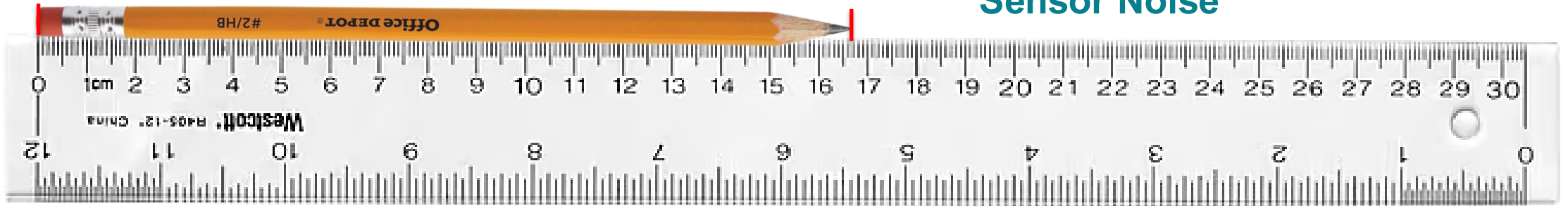
Mapping uncertainty

Where is this landmark?



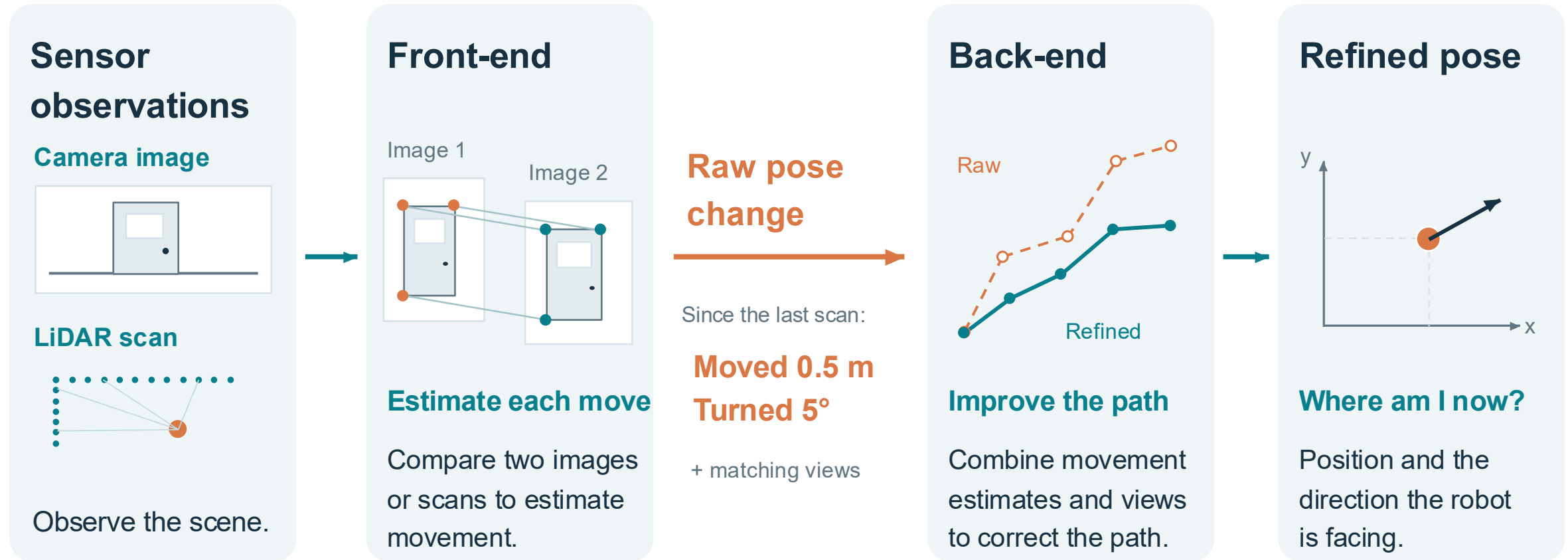
Pose uncertainty + sensor noise spread the landmark's map position.

Sensor Noise



From Sensor Measurement to Robot Pose

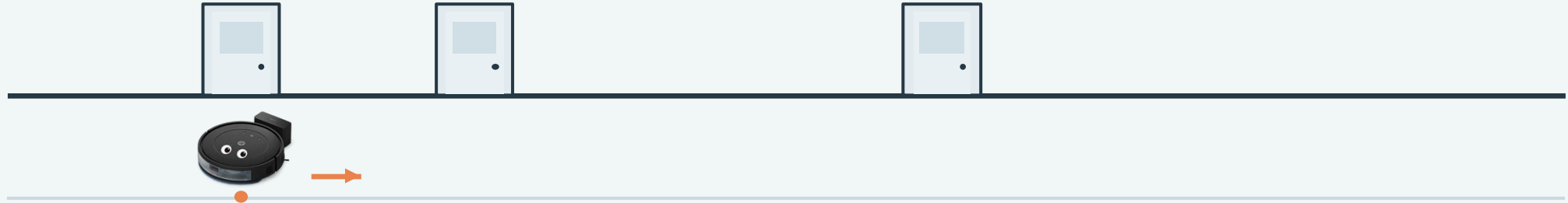
Expand the three-robot localization example into 2D planar space



Have we seen this place before?

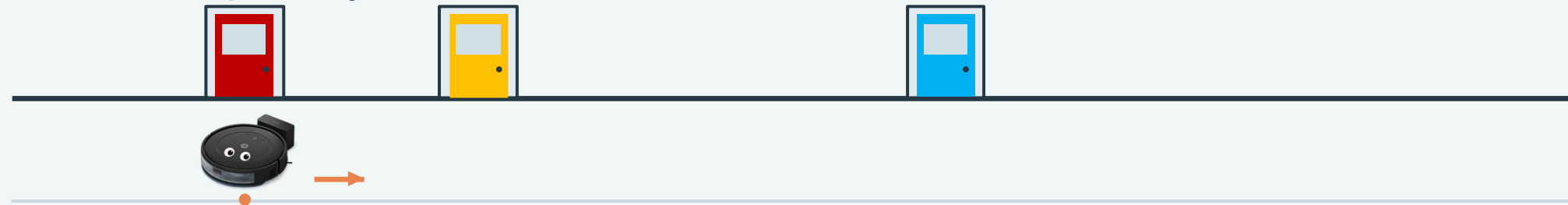
Similar-looking doors

Classroom corridor



Is this the same door, or another similar door?

Distinctive features help identify the door



Data association: match an observed door to one seen earlier.

Loop closure: verify a revisit and connect the two robot poses.

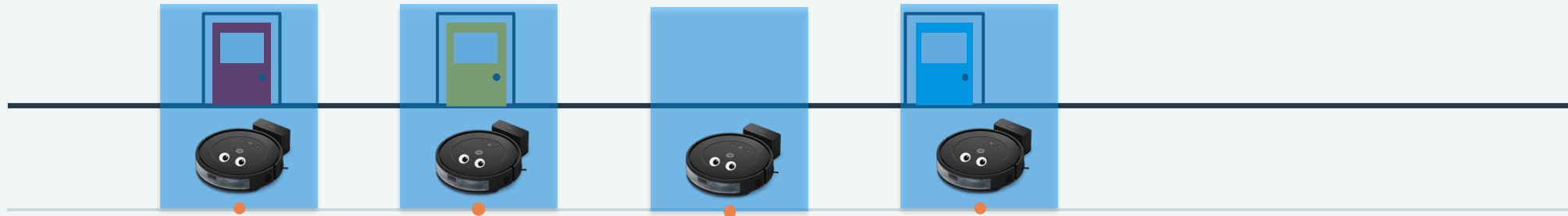
Q: How does the robot predict its position?

A: Use a motion estimate.

Start at the first door



Predicted positions: constant-speed assumption

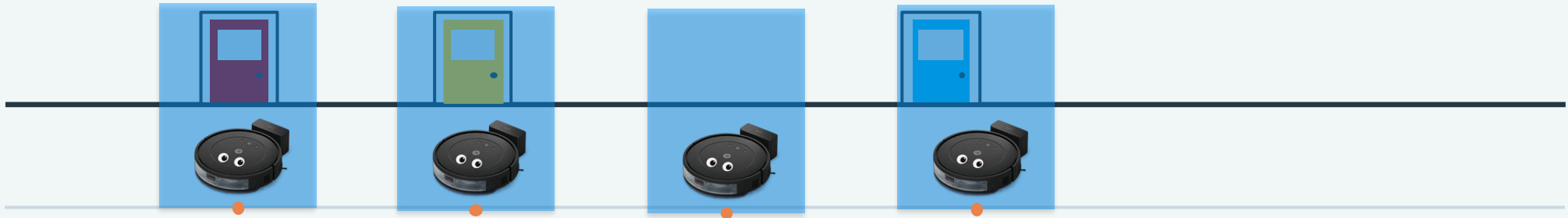


For this example, assume constant speed between observations.

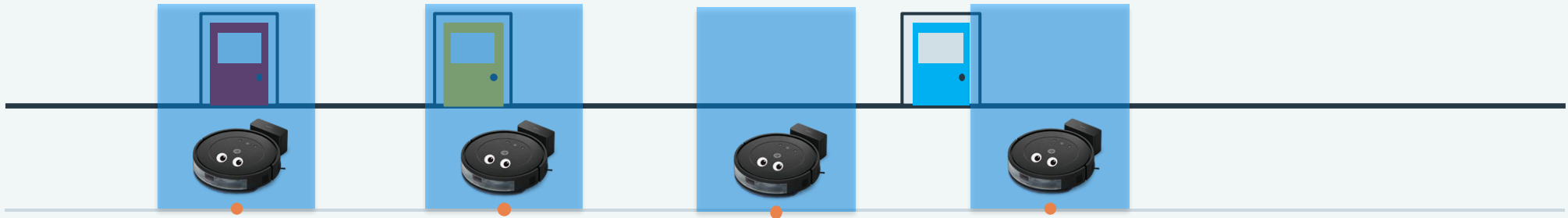
Q: What if the robot moves differently?

A: Position error accumulates.

Predicted positions: constant-speed assumption



Actual positions: the robot travels farther



Here, the robot travels farther than predicted; the doors stay fixed.

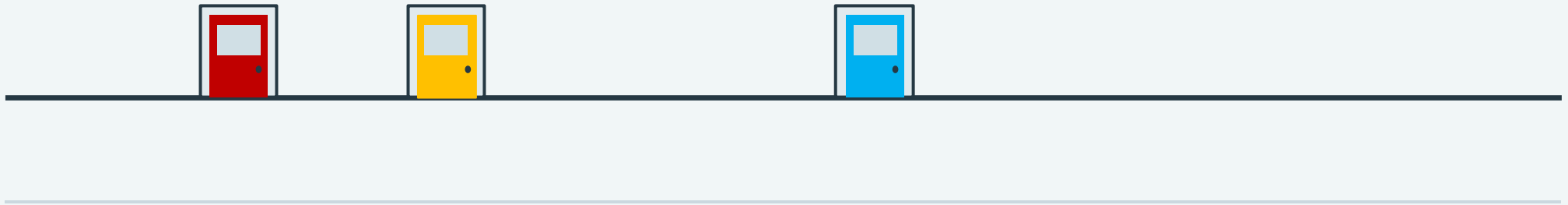
Q: How does position error affect the map?

A: Doors are mapped in the wrong places.

Actual door positions



Door positions in the robot's estimated map

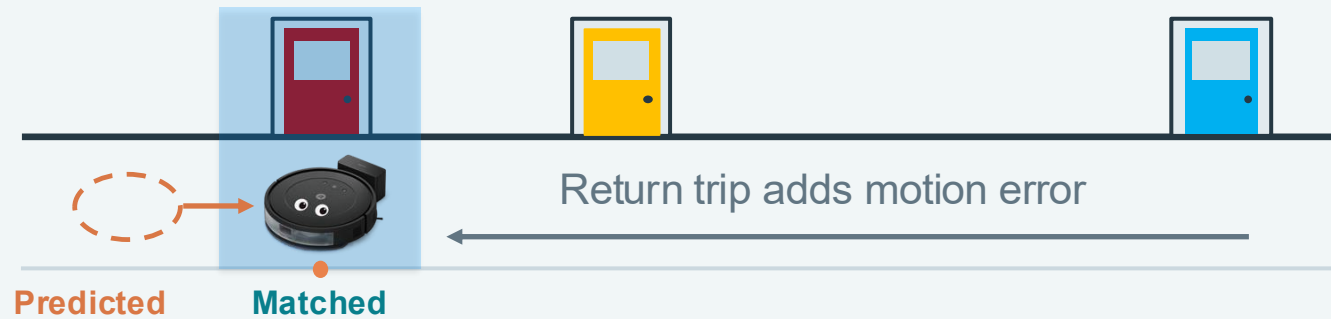


Underestimated travel makes the doors appear too close together in the map.

Q: How can the robot reduce this error?

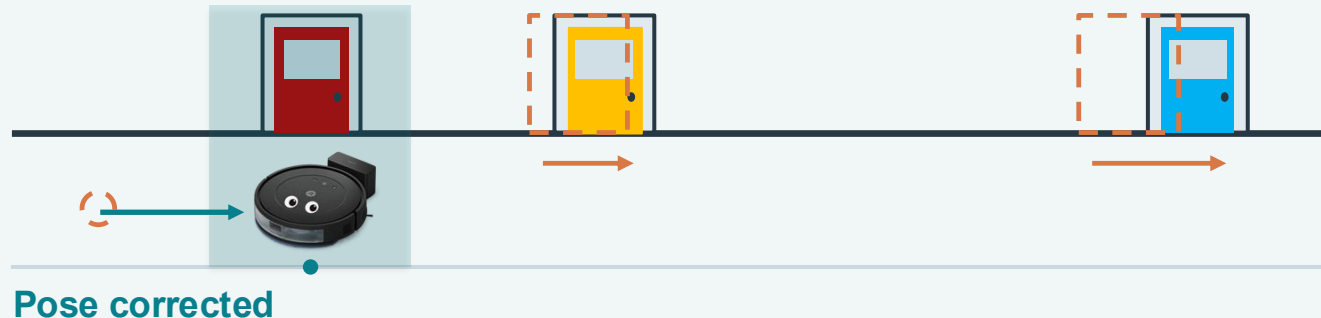
A: Revisit a familiar place!

1. Return to the first door and match the earlier scene



Recognize the door and nearby features from the first visit.

2. Correct the robot pose and refine the map



Combine both visits with the motion estimates.

--- Previous estimate

A verified revisit reduces error in the robot trajectory and the map.

Illustrative correction

What changes when the robot enters the water?

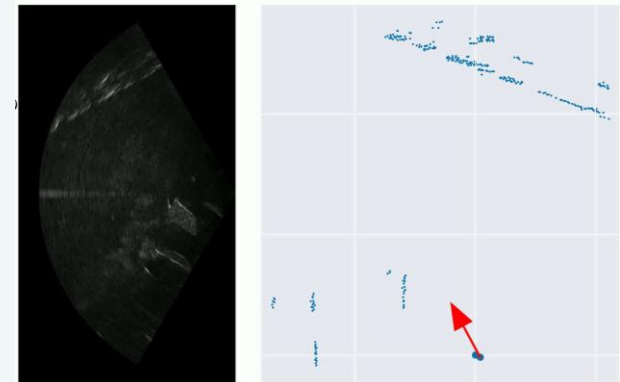
- **Autonomous Car (Waymo)**
- High-precision GPS + HD maps
- Mostly planar motion
- Vision and LiDAR fusion
- Rich, dense semantic structures



- **Autonomous Surface Vehicles (ASV)**
- GPS available but **degraded**
- Drift from **currents** and **wind**
- Vision and LiDAR fusion
- **Uneven** structures



- **Underwater Unmanned Vehicles (UUV)**
- **No** GPS
- Influenced by **hydro-dynamics**
- Only sonar in murky water
- **Unstructured**



Marine sensors: what they do in the field

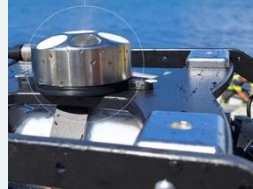
Each sensor provides a different piece of the robot's position and surroundings.

IMU



Rotation + acceleration
Predict short-term motion

DVL



Velocity over the seabed
Needs seabed echoes

Pressure sensor



Pressure → depth
Constrain vertical motion

Position fixes



GNSS: at the surface
Acoustics: underwater

Cameras: match visual features



GoPro on the field rig

Waterproof housing
Clear water + light → observe visible features



Blue Robotics camera

Inside a pressure housing

Sonar: match acoustic structure



Oculus 750d · 2D



Sonar 3D-15 · 3D

Murky water → observe walls and objects

Motion and depth aid localization; cameras and sonar reveal the environment.

Perception underwater: cameras and sonar

Water clarity changes what the robot can observe.

GoPro Camera Clear water

- Rock texture and edges remain visible.



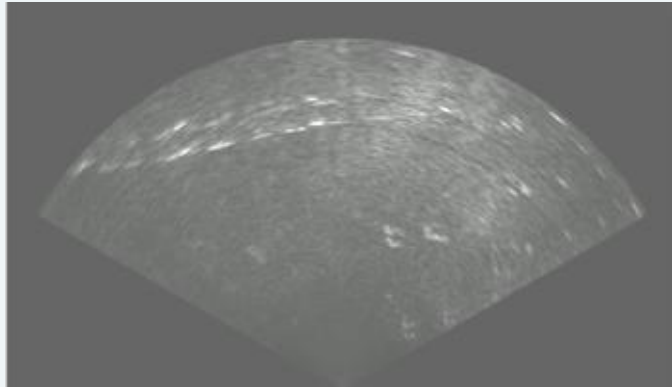
GoPro Camera Murky water

- Haze reduces contrast and hides texture.



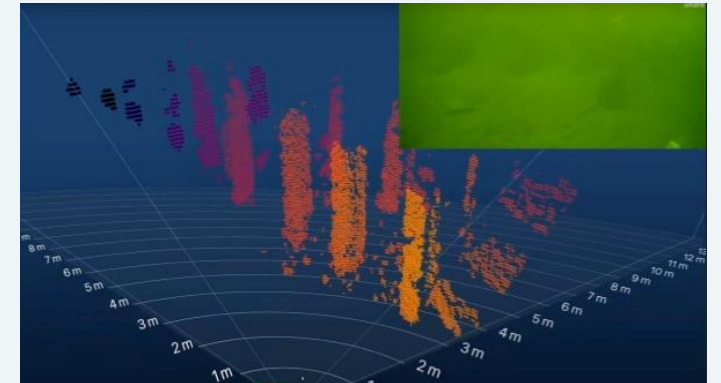
Oculus 750d Imaging sonar

- Noisy but distinguishable.
- Max range: 120m
- Resolution: 0.1m



Water Linked Sonar 3D-15

- Denser cloud
- Max range: 15m
- Resolution: 1.5mm



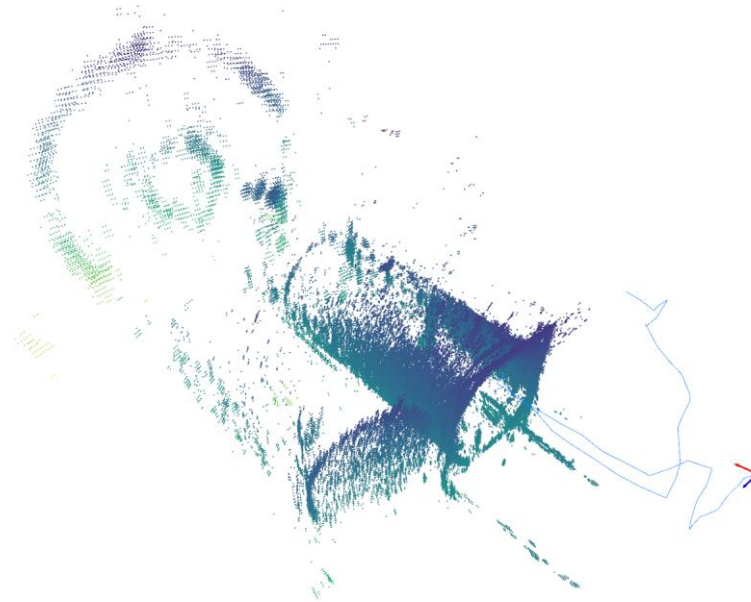
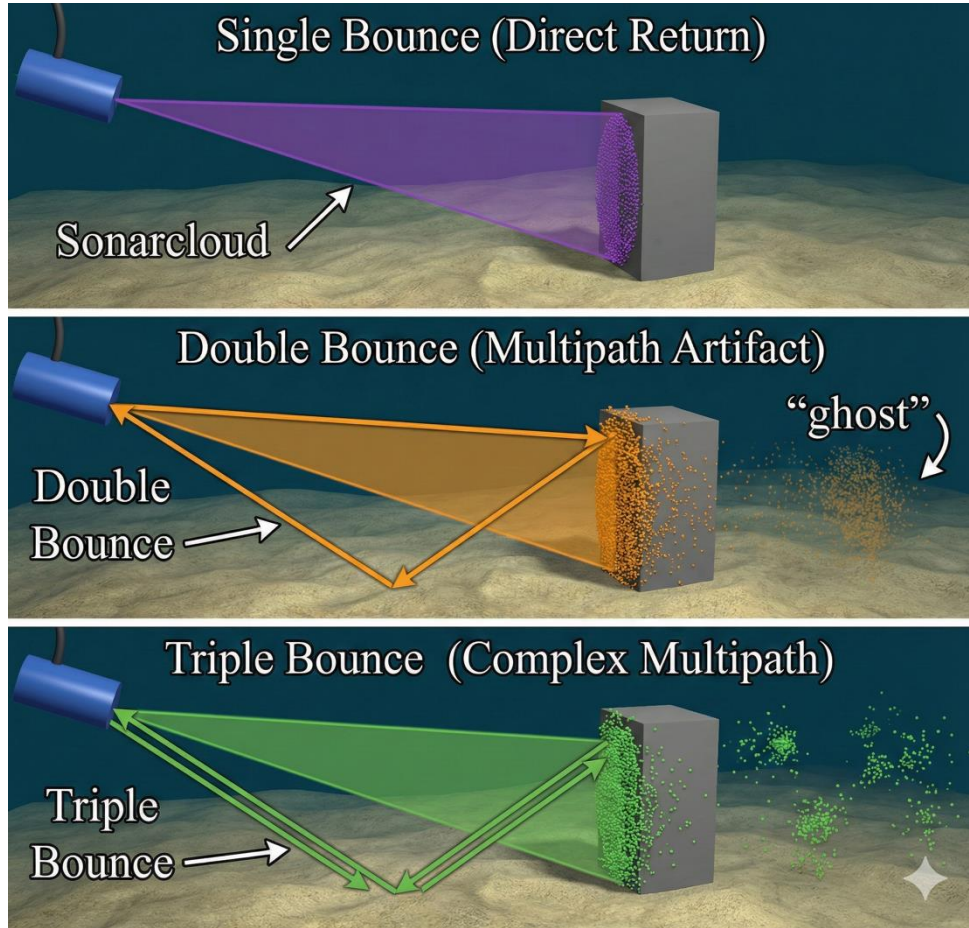
As visibility drops, sonar can still reveal structure.

Sonar multipath: one object, multiple echoes

Longer paths create delayed echoes

Reflections add extra travel time.

Delayed echoes can appear as false points.



Ghost structure in a sonar map

Reduce the impact

1 Front-end

Filter unreliable returns.

2 Sensor fusion

Combine sensors to reduce pose uncertainty.

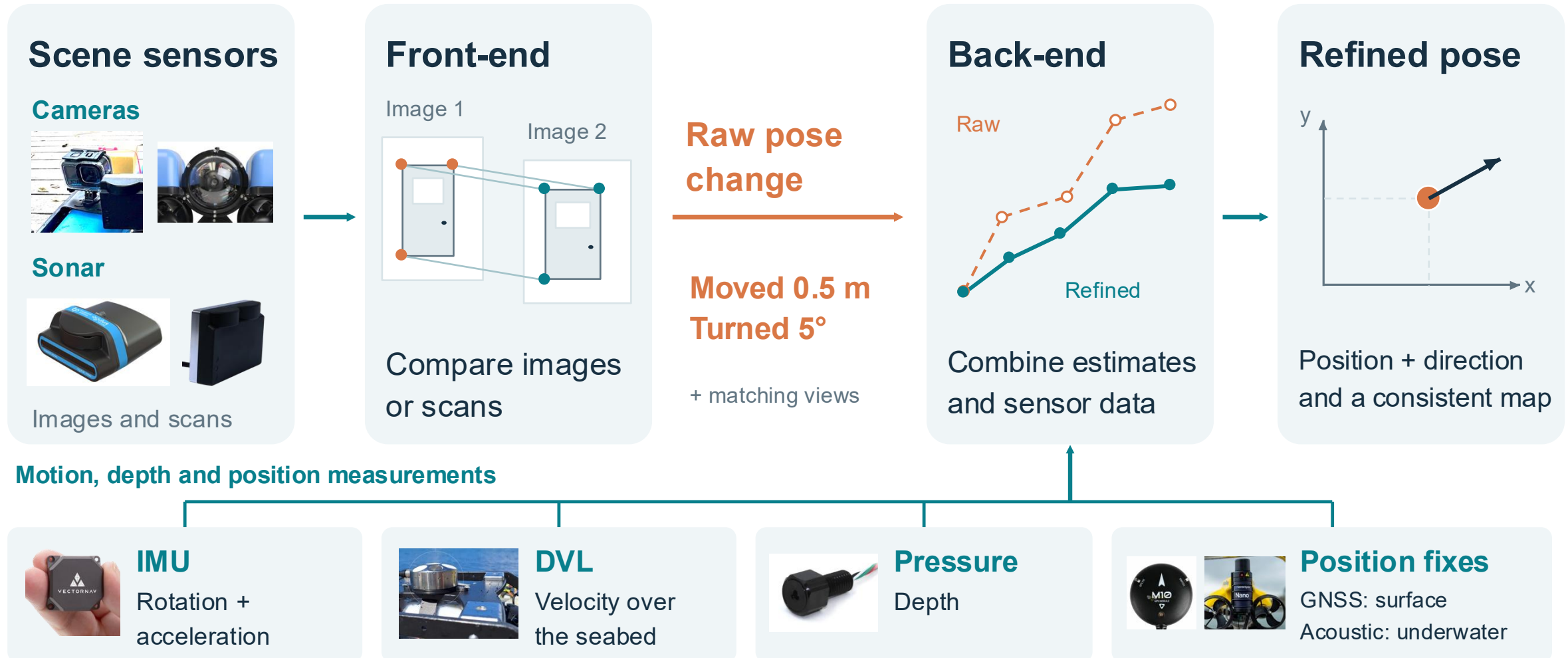


Camera view

False points can create incorrect matches and distort the map.

Multimodal fusion: combine complementary sensors

The same pipeline, with motion, depth and position measurements added.



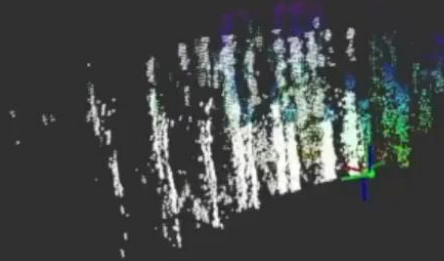
Fuse complementary measurements, giving reliable readings more weight.

From 3D sonar scan to dense point-cloud map

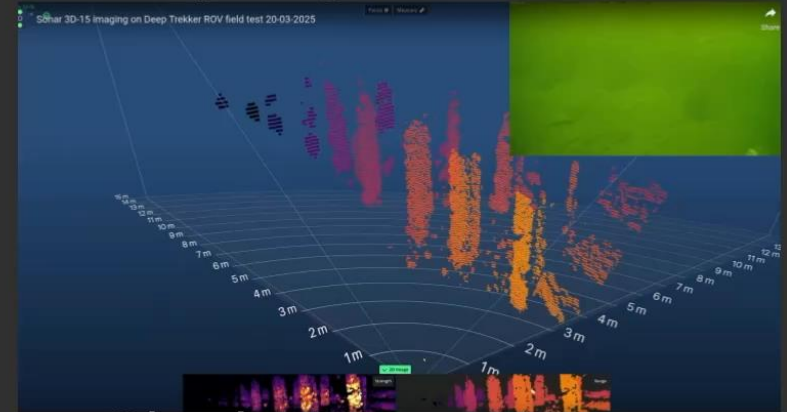
Sonar registration in pier piling environment



Pier piling environment

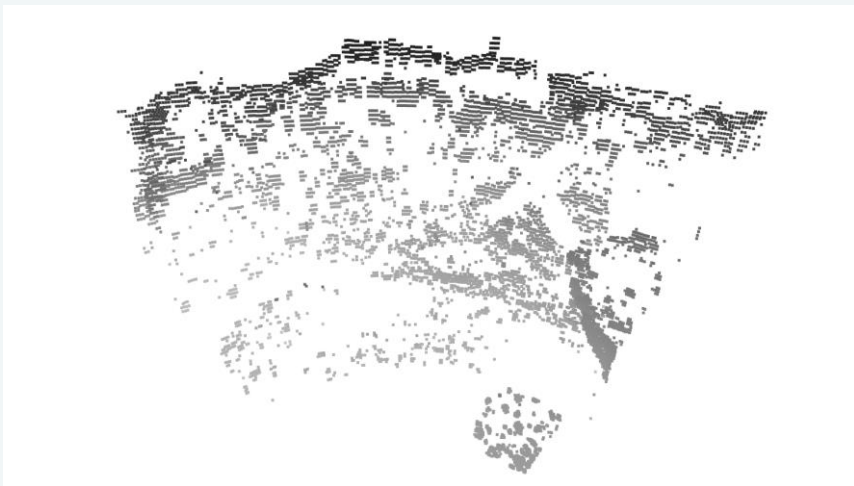
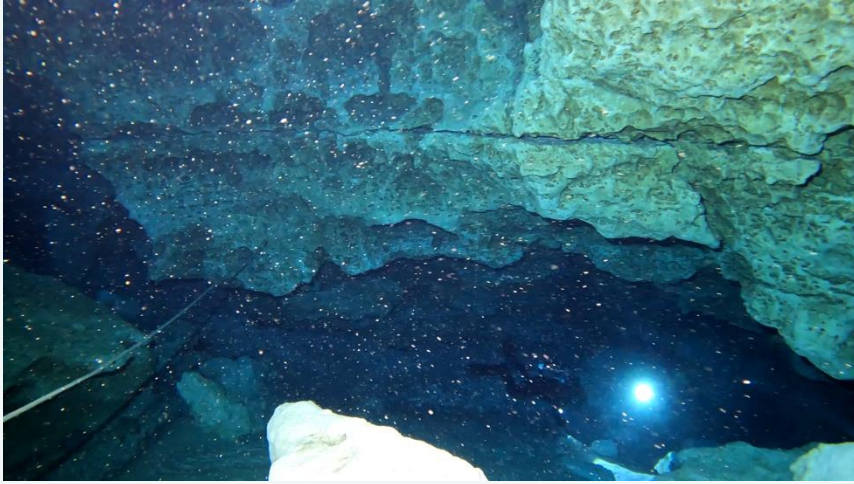


2x

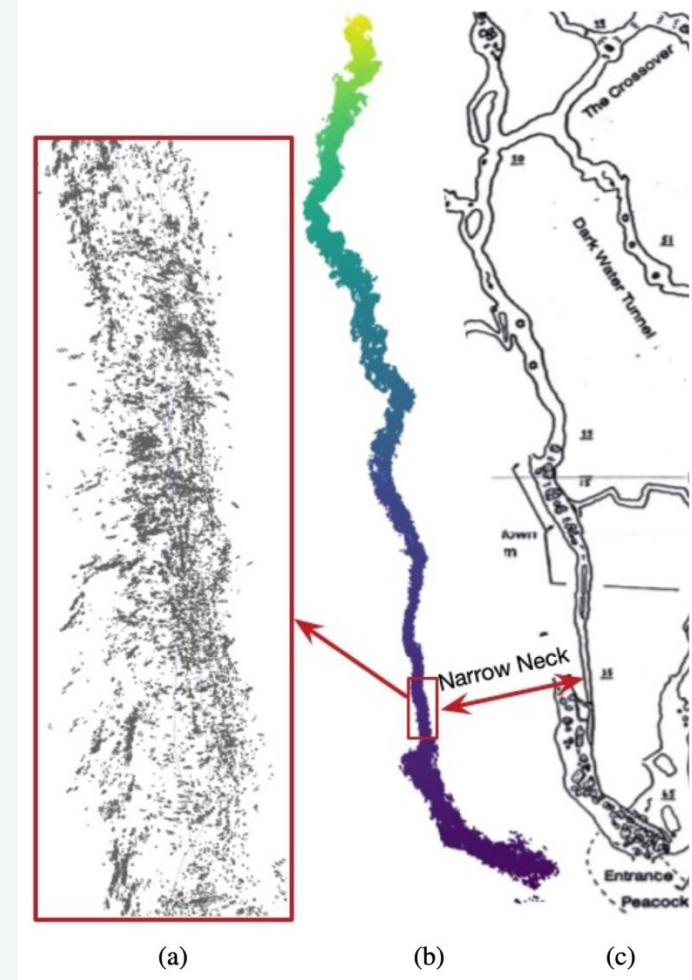


Single 3D sonar scan
& camera visibility

Results



Ginnie Spring, FL



Peacock Cave, FL

Serves as a spatial reference for water quality mapping within the cave.

Additional challenge for multi-robot case

Underwater Unmanned Vehicles (UUV)

- **No** GPS (Seawater strongly attenuates radio waves)
- Influenced by **hydro-dynamics** (robot would rotate/drift)
- Only sonar in murky water
- **Unstructured** Features



Multi UUV

- Even high-speed(hs) acoustic modems are limited to **62.5 kbps** (Wi-Fi at 10–1000 Mbps).
- Rendezvous is not allowed due to **acoustic interference from identical-frequency sonars**

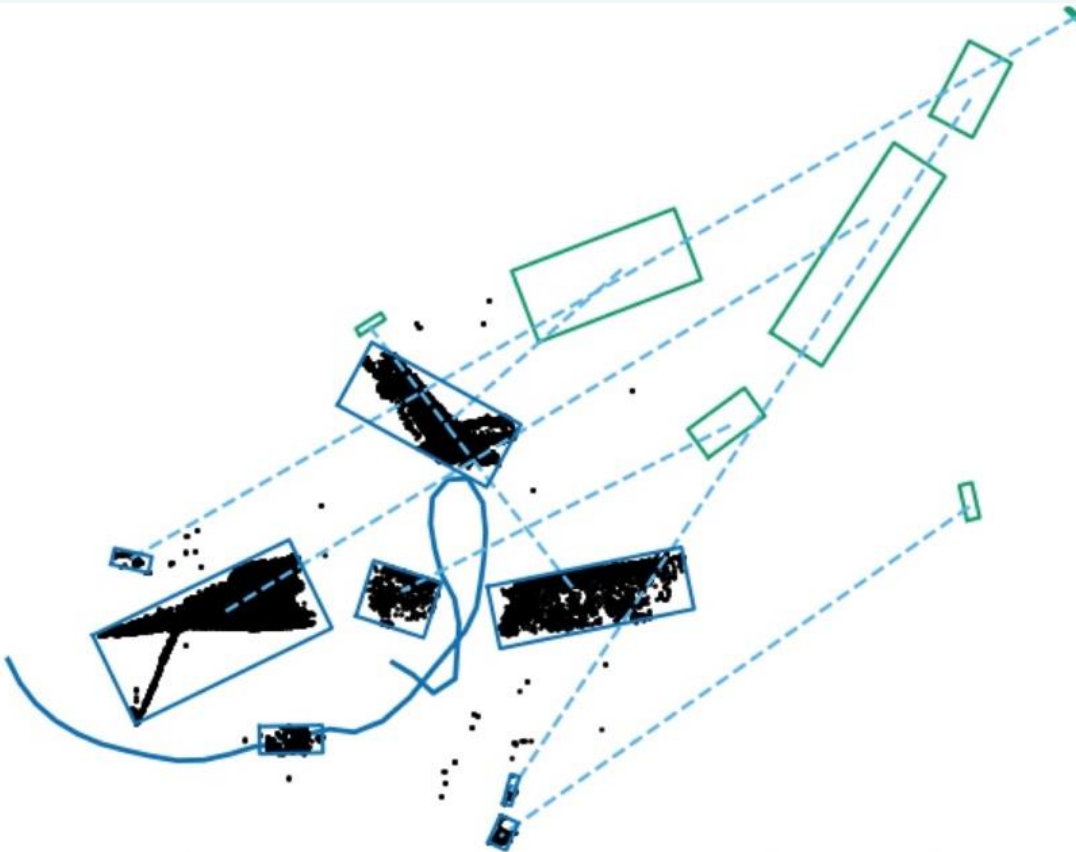


<https://www.evologics.com/acoustic-modems>

EvoLogics underwater acoustic modems.

Underwater teams must share information carefully

Object graph matching for inter robot data association



Objects

Bounding boxes describe the size of each cluster.

Relations

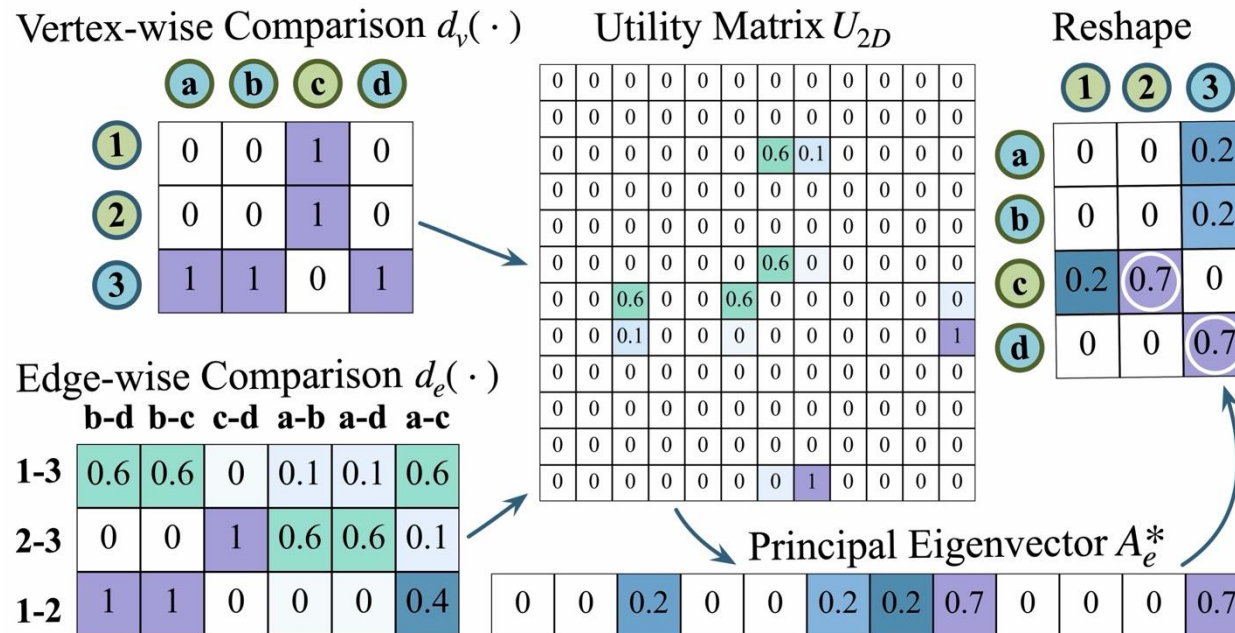
Distances between centers describe the scene geometry.

Correspondences

Match the two object graphs to estimate relative alignment.

Use shared geometric structure when the relative robot poses are unknown.

Use graph matches to initialize scan alignment



Object-graph matching process

1 Match object graphs

Find corresponding objects.

2 Estimate an initial transform

Bring the two maps into alignment.

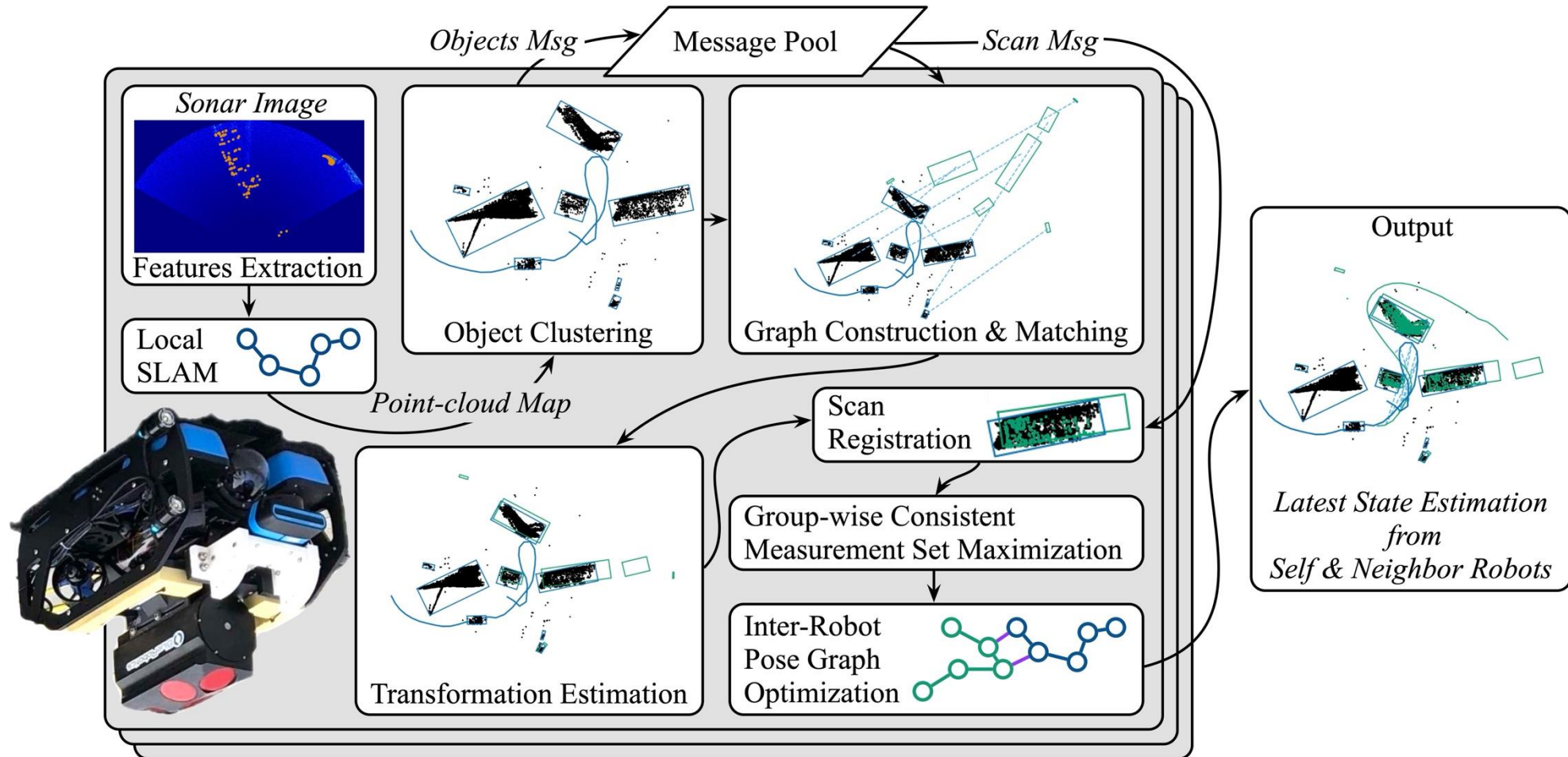
3 Refine with ICP

Align the overlapping sonar scans.

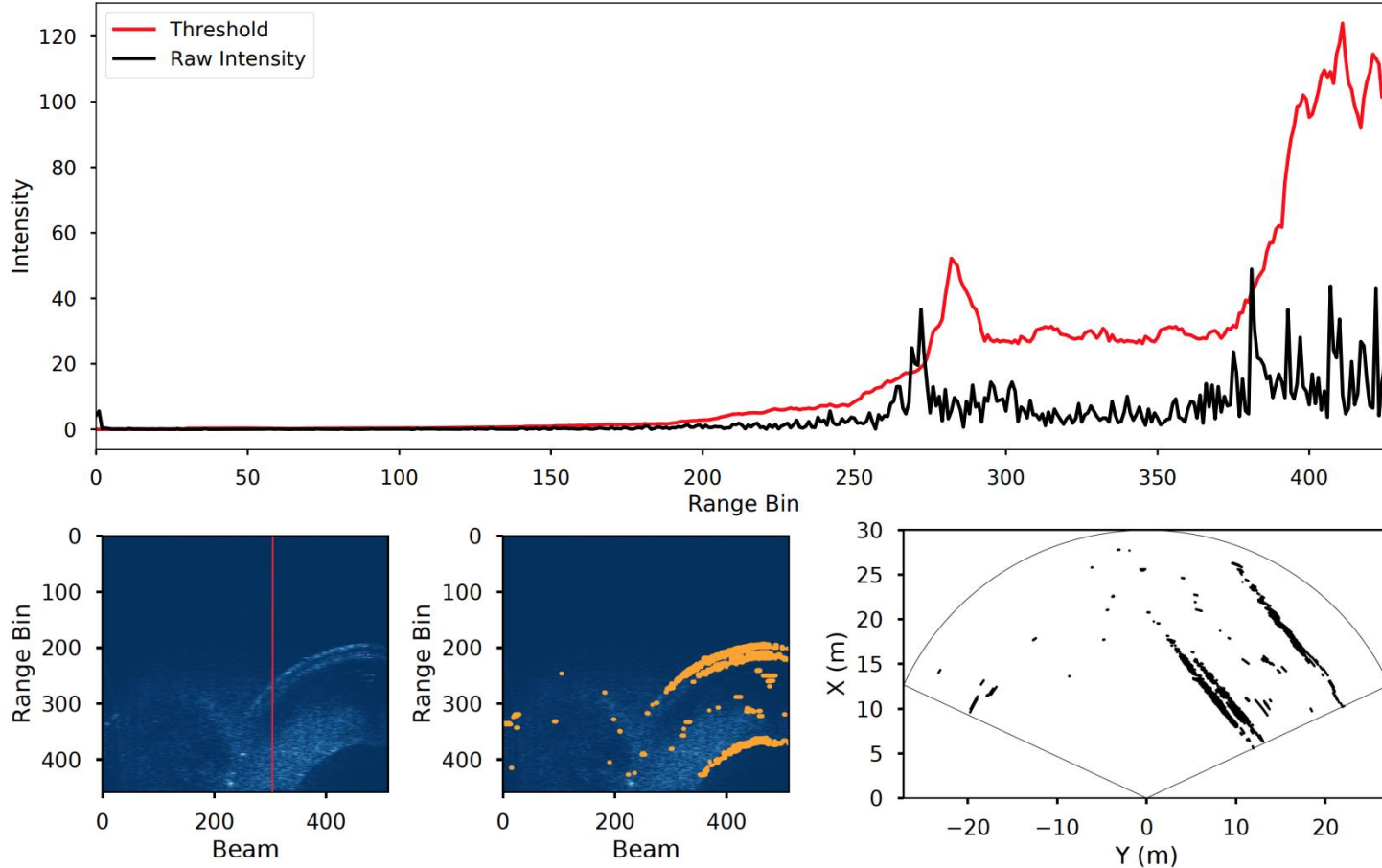
Mean registration time per step: 10.11 s (DRACo1) → 1.55 s (DRACo2)

DRACo-SLAM2: from sonar to a shared map

Local maps → object matching → loop closures → pose graph optimization



Extract structure from a noisy sonar image



Estimate local noise

Use nearby returns to estimate the background.

Keep strong returns

Select returns above a local threshold.

Create feature points

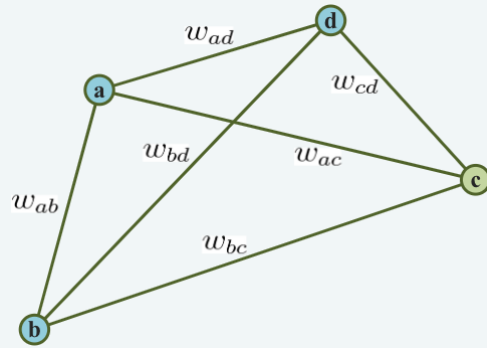
Convert range and bearing to Cartesian points.

SOCA-CFAR feature extraction · Oculus M750d imaging sonar

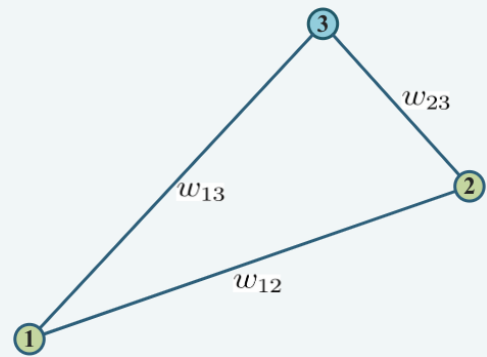
Local thresholds adapt as the acoustic background changes.

Use object size and geometry to find a match

Local objects

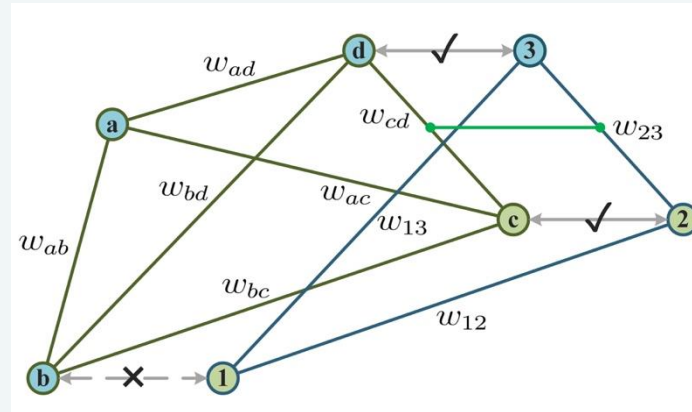


Neighbor objects

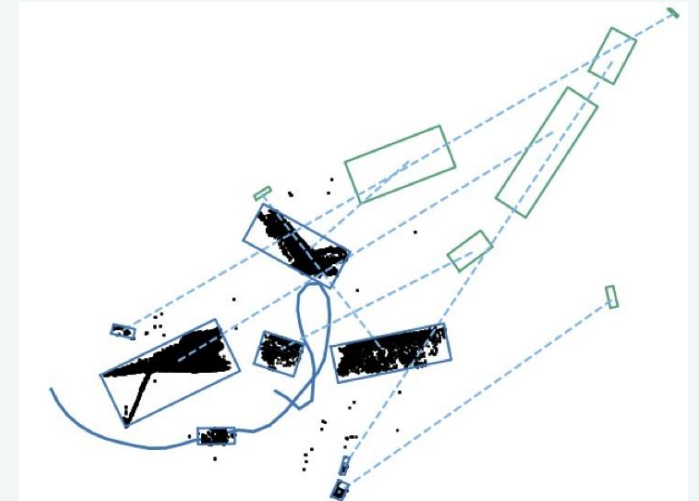


Compare geometry

Object dimensions
Distances between centers



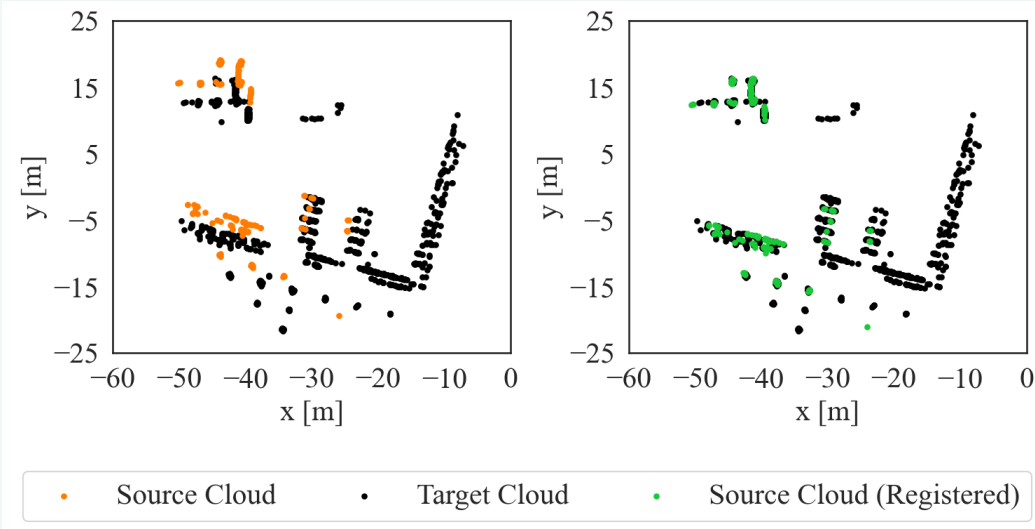
Matched objects



DBSCAN clusters the local point cloud into objects before matching.

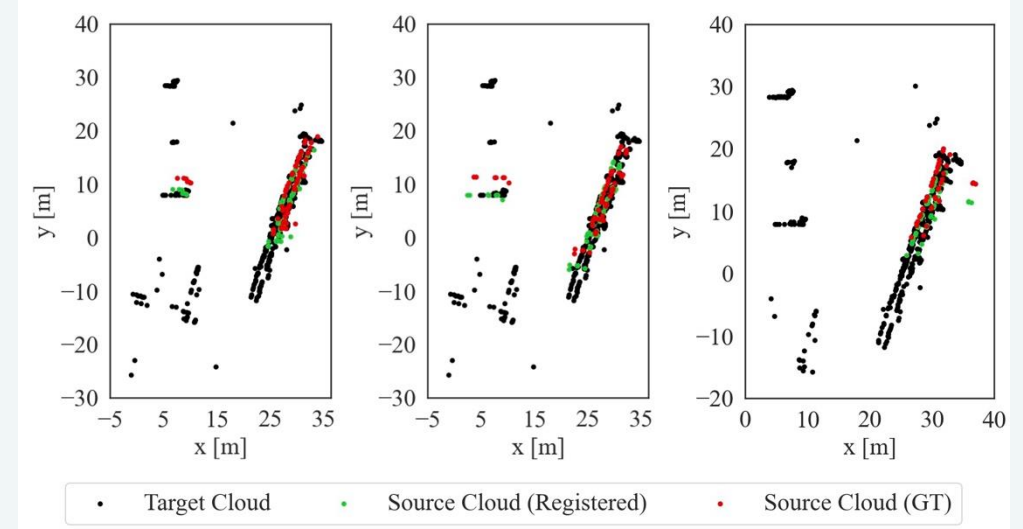
Check loop closures across the robot group

A useful initial alignment



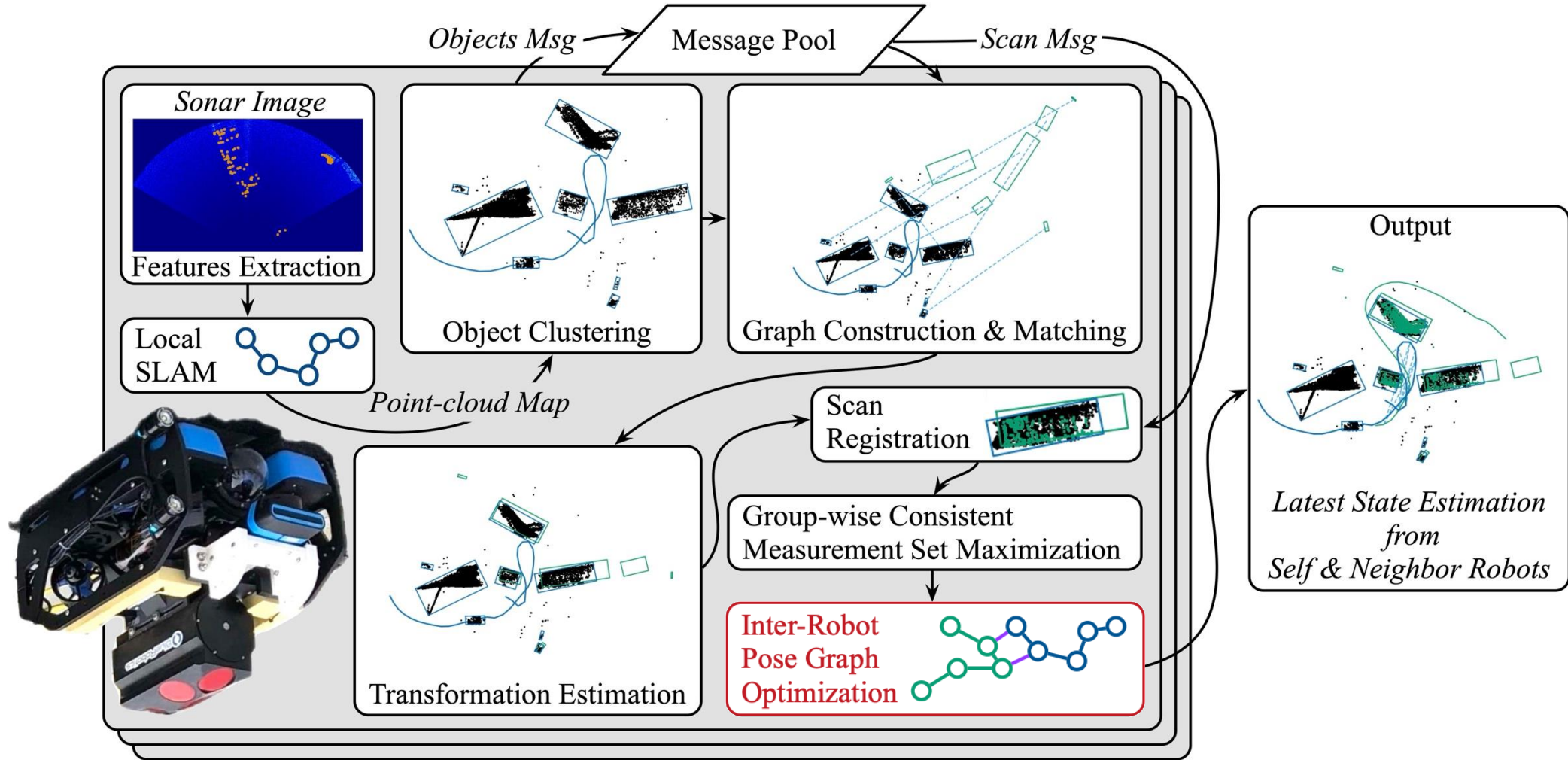
Graph matching gives ICP
a useful starting pose.

Similar errors can repeat



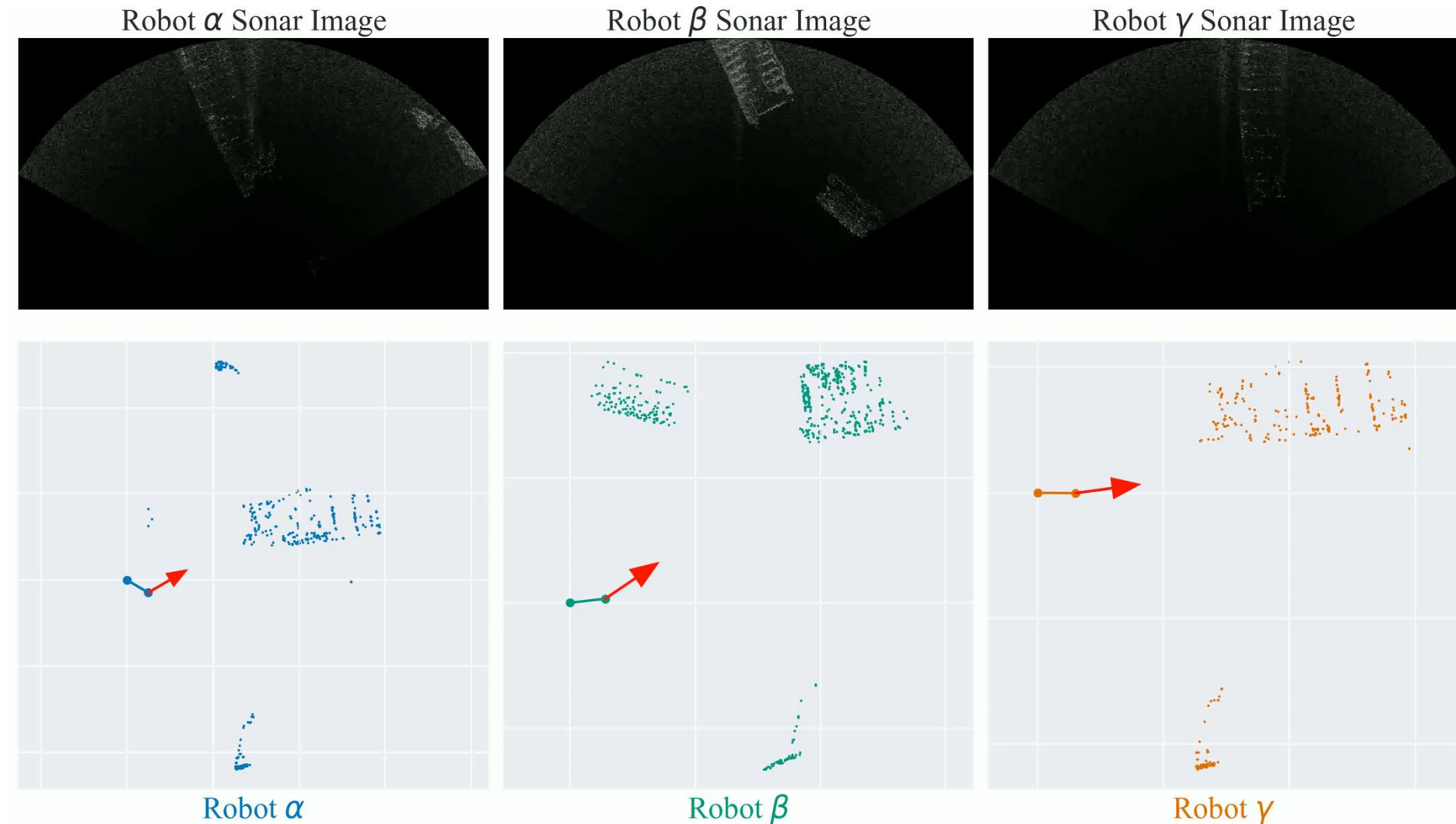
Nearby scan pairs can agree
with one another and still be wrong.

Verified matches improve the shared map



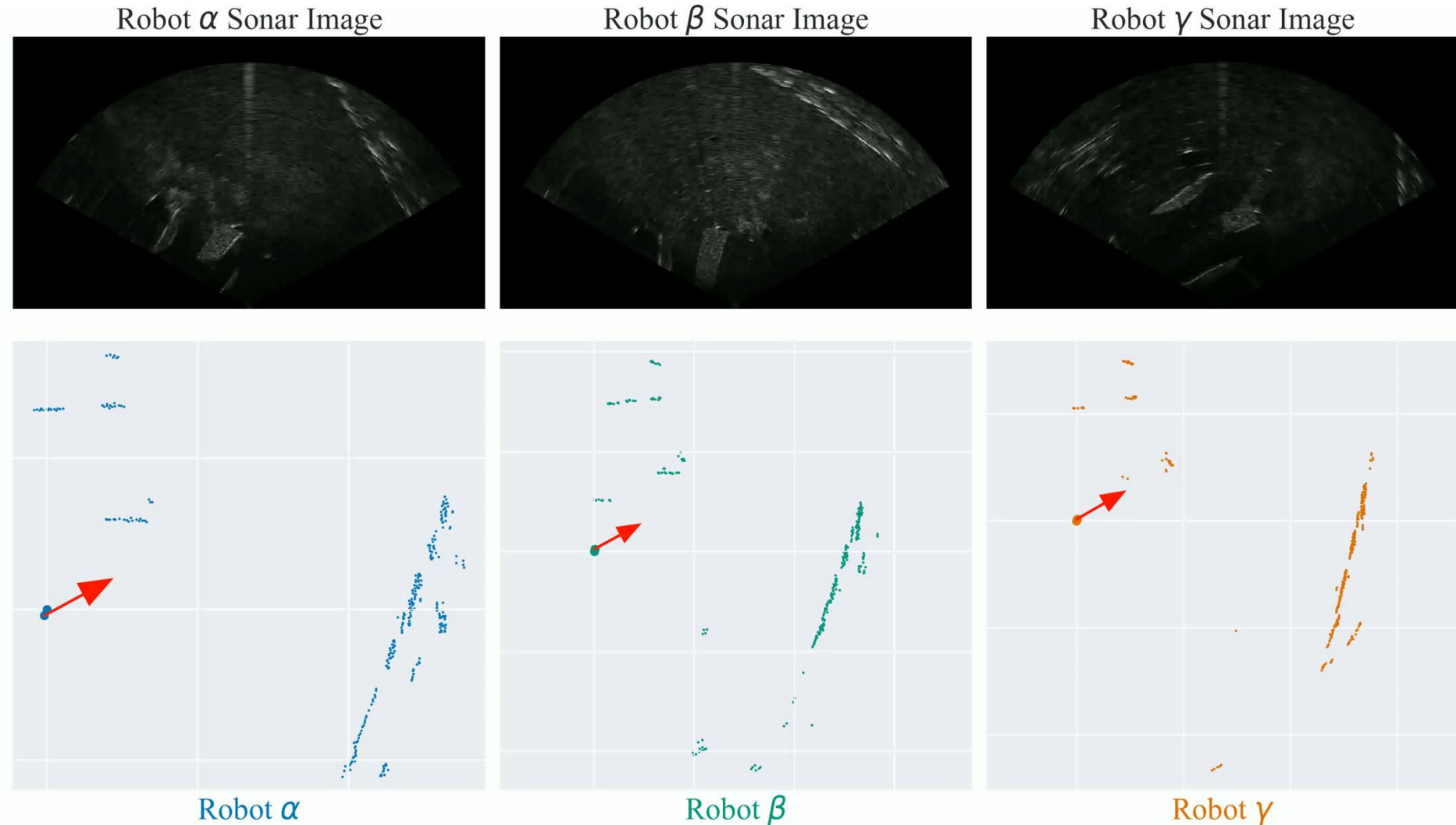
Accepted inter-robot loop closures become constraints in pose graph optimization.

Three-robot mapping in an airplane simulation



Simulated sonar observations and maps from three robot trajectories.

Mapping with real sonar data from USMMA



Recorded sonar sessions with simulated three-robot communication.

Autonomous exploration on the water

Explore efficiently while staying localized

A changing navigation problem

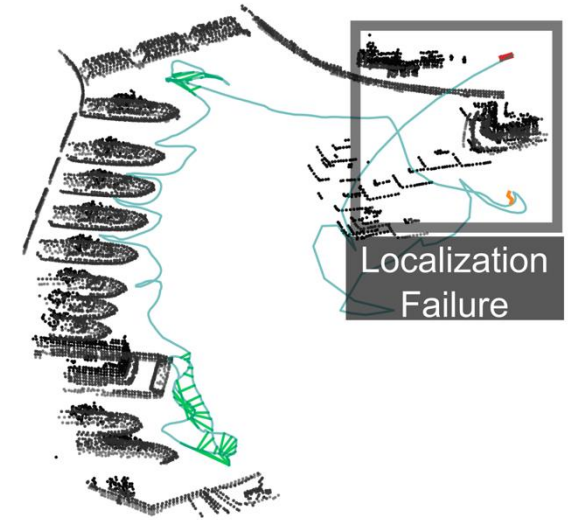
GNSS is available but degraded.
Currents and wind change the motion.
Useful landmarks vary along the shore.

Why explore autonomously?

The robot chooses its own path to discover an unknown environment.
Build a useful map while deciding where the robot should go next.



Perception, mapping, and environmental sensing



Localization drift can distort the map

What should a good route do?

Discover new areas

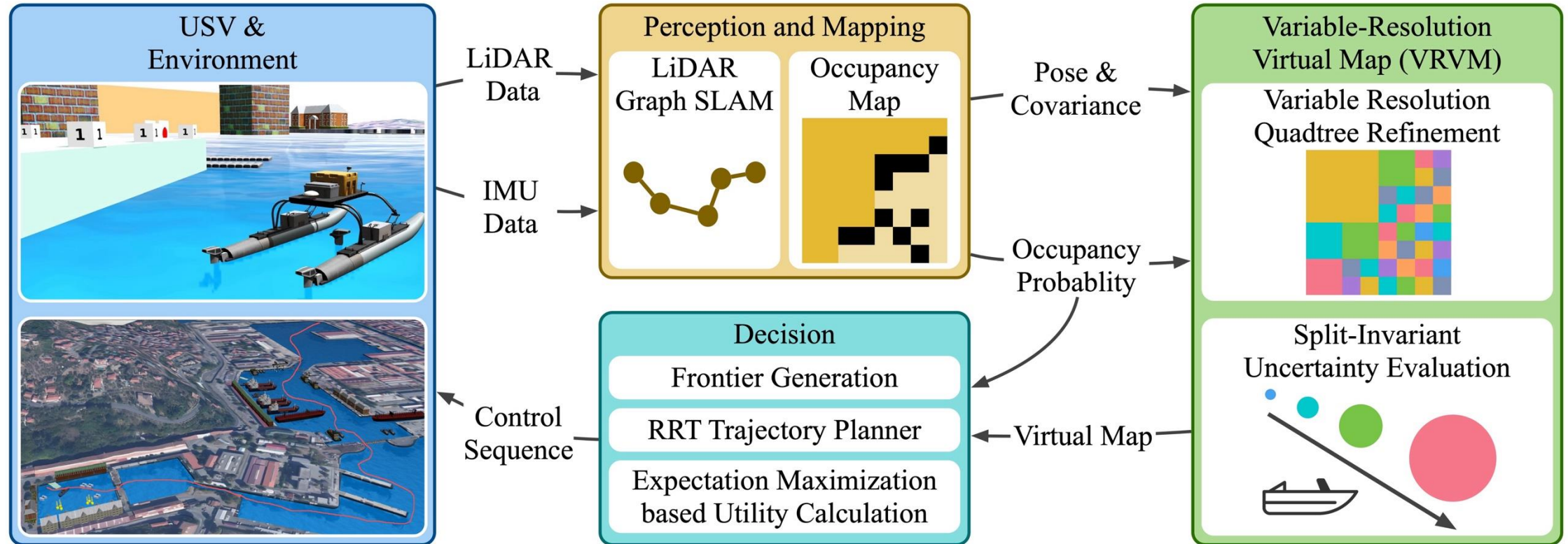
Keep localization and mapping reliable

Limited energy and onboard computation

Use limited time and computation well

The exploration loop

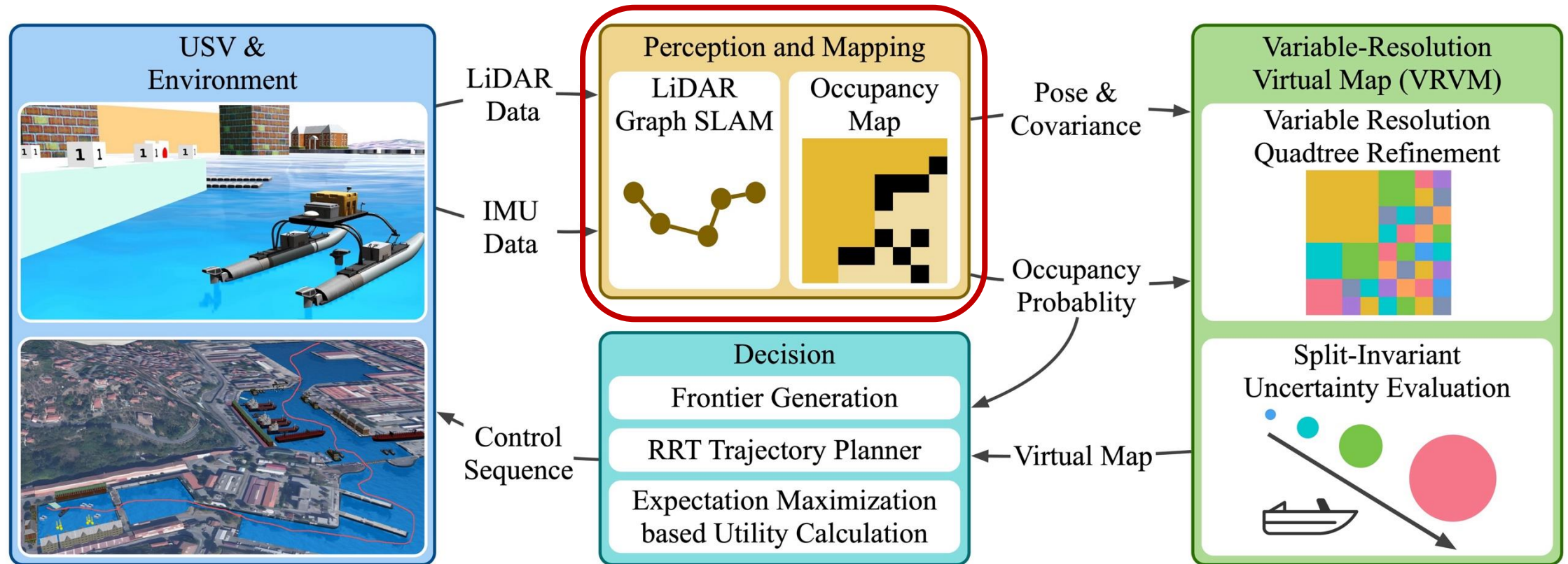
Sense the environment, update the map, and choose the next route.



A new observation starts the next planning cycle.

Step 1: Localize and build the map

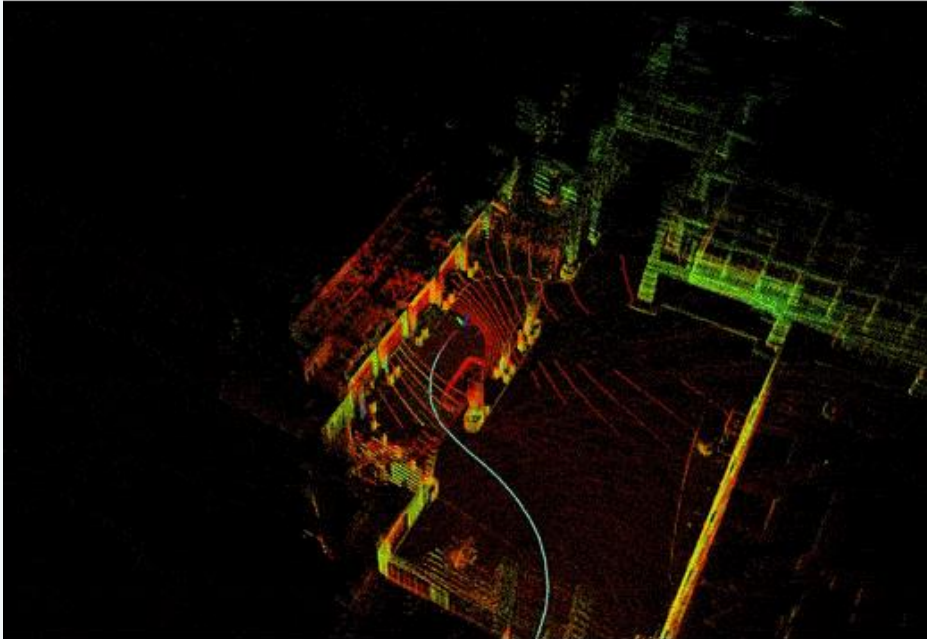
LiDAR and IMU measurements estimate motion and reveal obstacles.



The planner needs both a map and an estimate of its uncertainty.

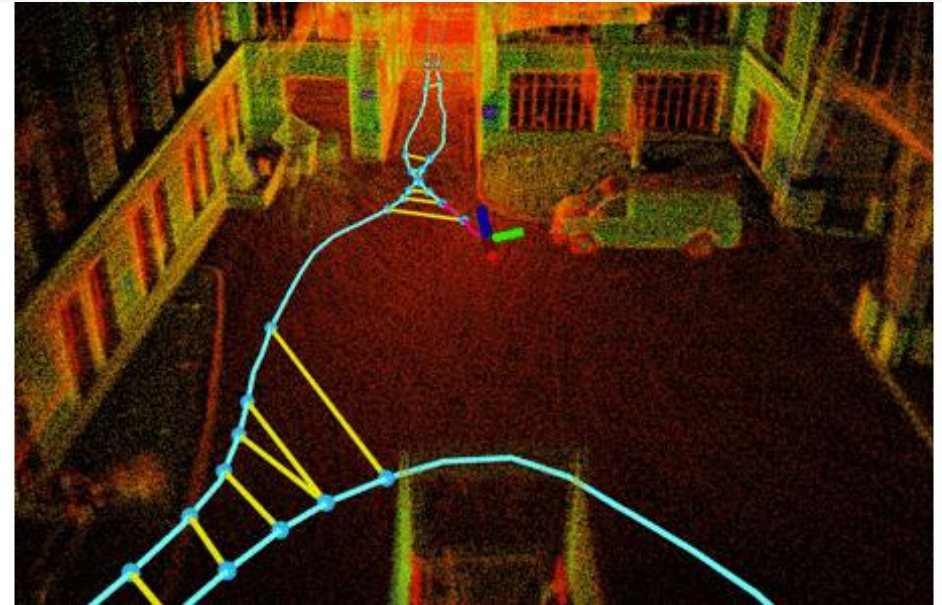
LiDAR observations become motion constraints

Match consecutive scans



Estimate the robot's motion

Recognize a revisit

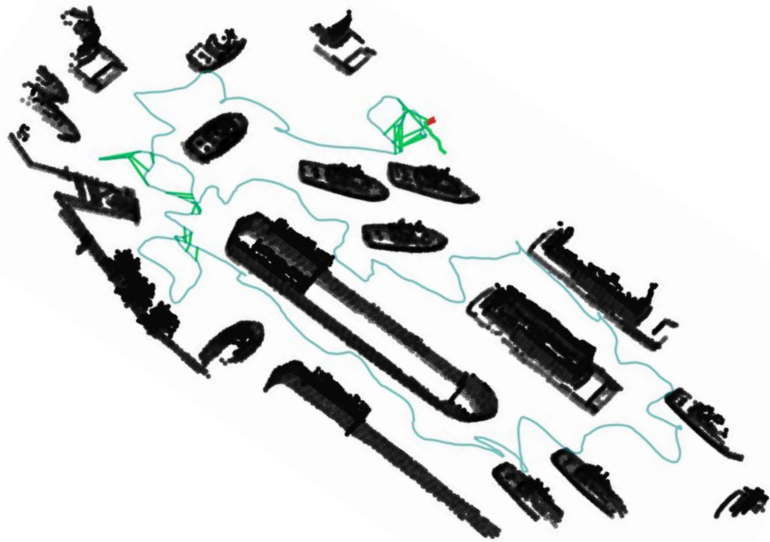


Loop closure + pose graph optimization

An occupancy map shows where the robot can go

Occupancy Map: a map that indicates which areas are **free, occupied, or unknown**, based on sensor observations.

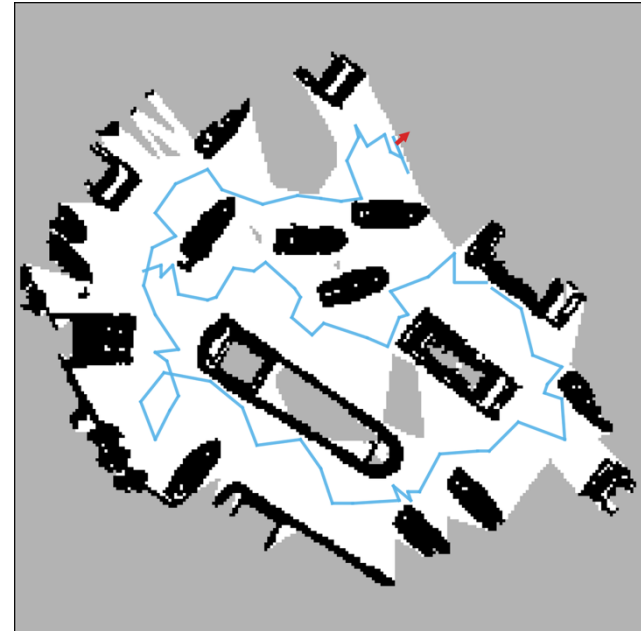
Localized observations



Geometry + estimated robot path



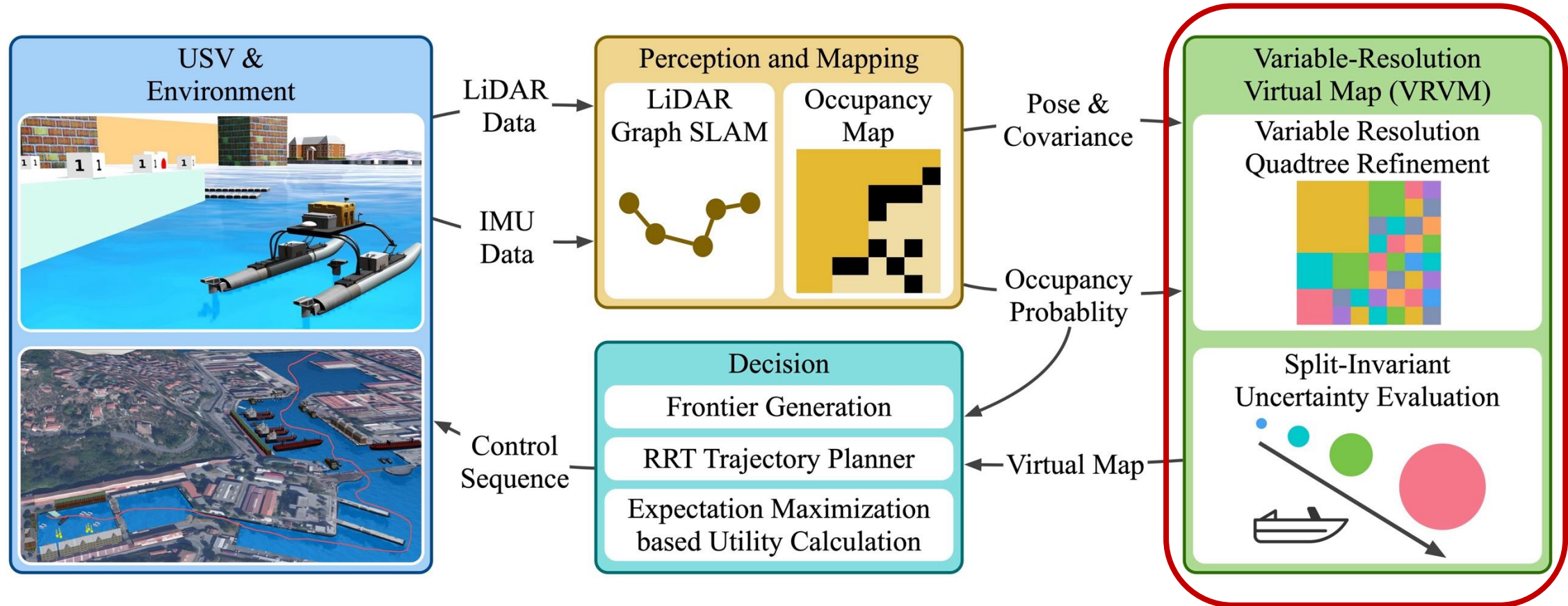
Occupancy grid



Free / occupied / unknown

Step 2: Represent uncertainty in a virtual map

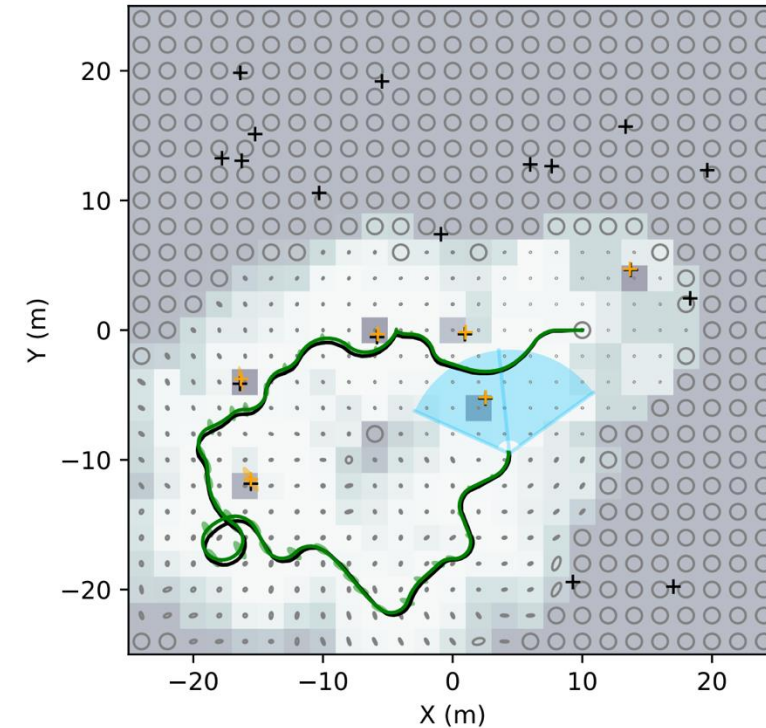
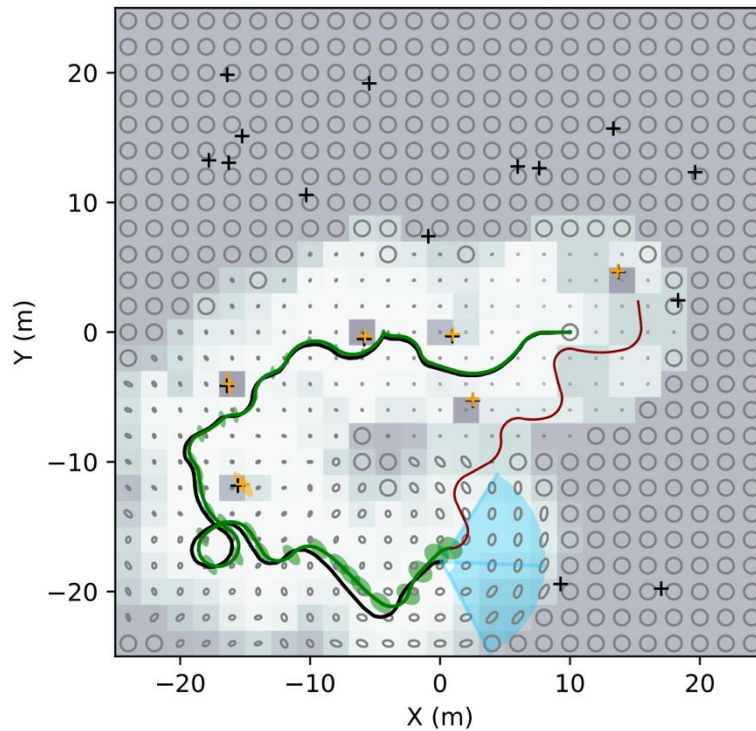
A virtual map describes where the robot expects reliable information.



Use the current belief to evaluate possible future routes.

A virtual map predicts the value of a route

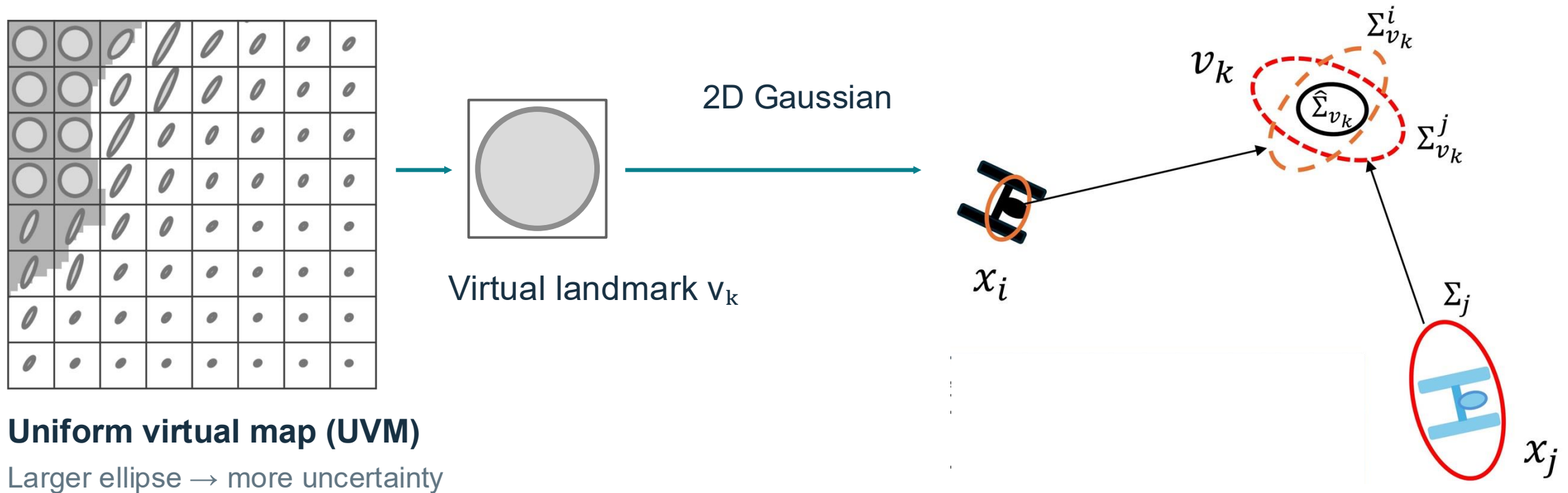
Virtual landmarks represent uncertainty about localization and the map.



Predict uncertainty along each route; balance information, localization, and travel.

Updating a standard virtual landmark

Predict uncertainty before moving, then combine estimates from different viewpoints.



1 Predict: belief propagation

$$\Sigma_{v_k}^i = A_k^i \tilde{\Sigma}_{x_i} A_k^{iT} + B_k^i \Lambda B_k^{iT}$$

Pose uncertainty + sensor uncertainty

2 Fuse: covariance intersection

$$\hat{\Sigma}_{v_k}^{-1} = \omega \Sigma_{v_k}^{i-1} + (1 - \omega) \Sigma_{v_k}^{j-1}$$

Combine estimates without assuming independence.

Use the predicted uncertainty to compare candidate routes.

Use fine resolution where it is useful

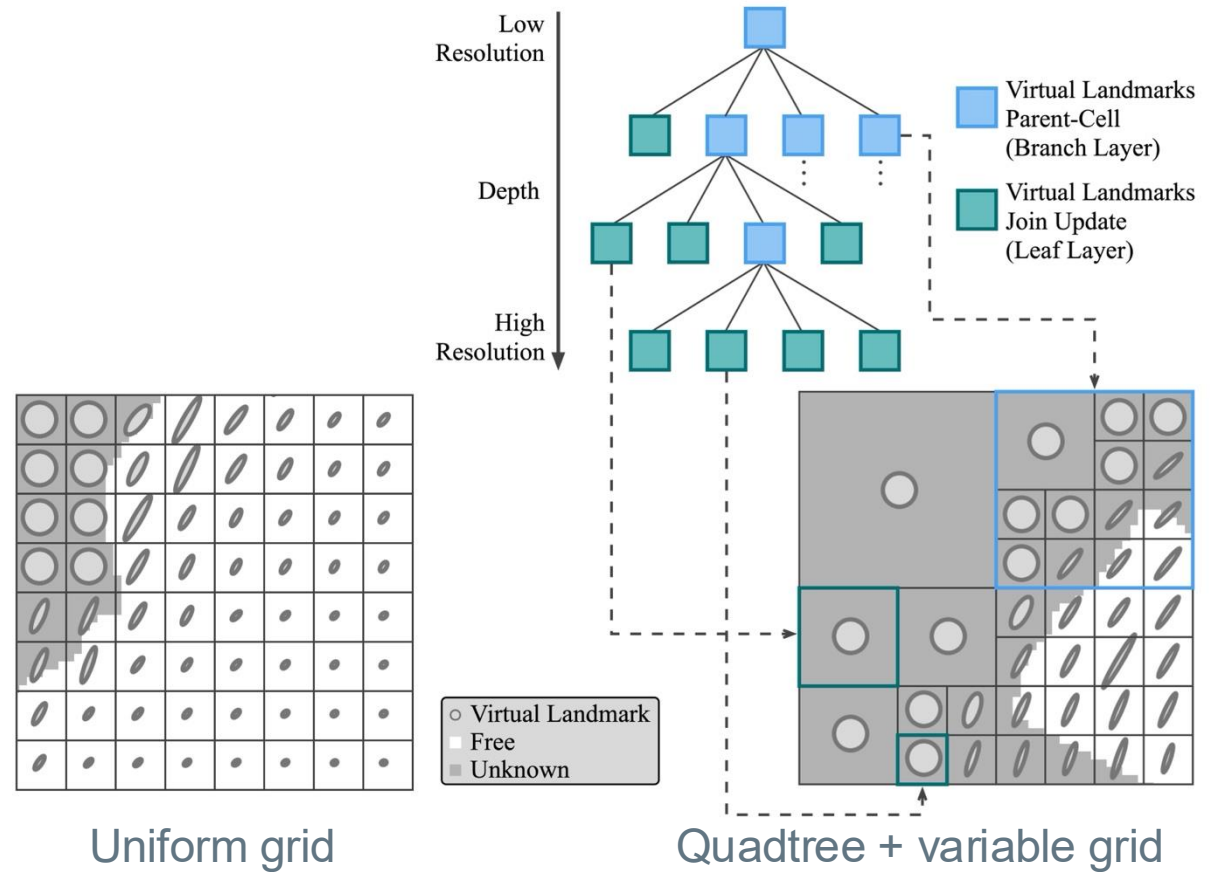
Uniform virtual map

- The same cell size everywhere.
- More cells increase the computation.

Variable-resolution virtual map

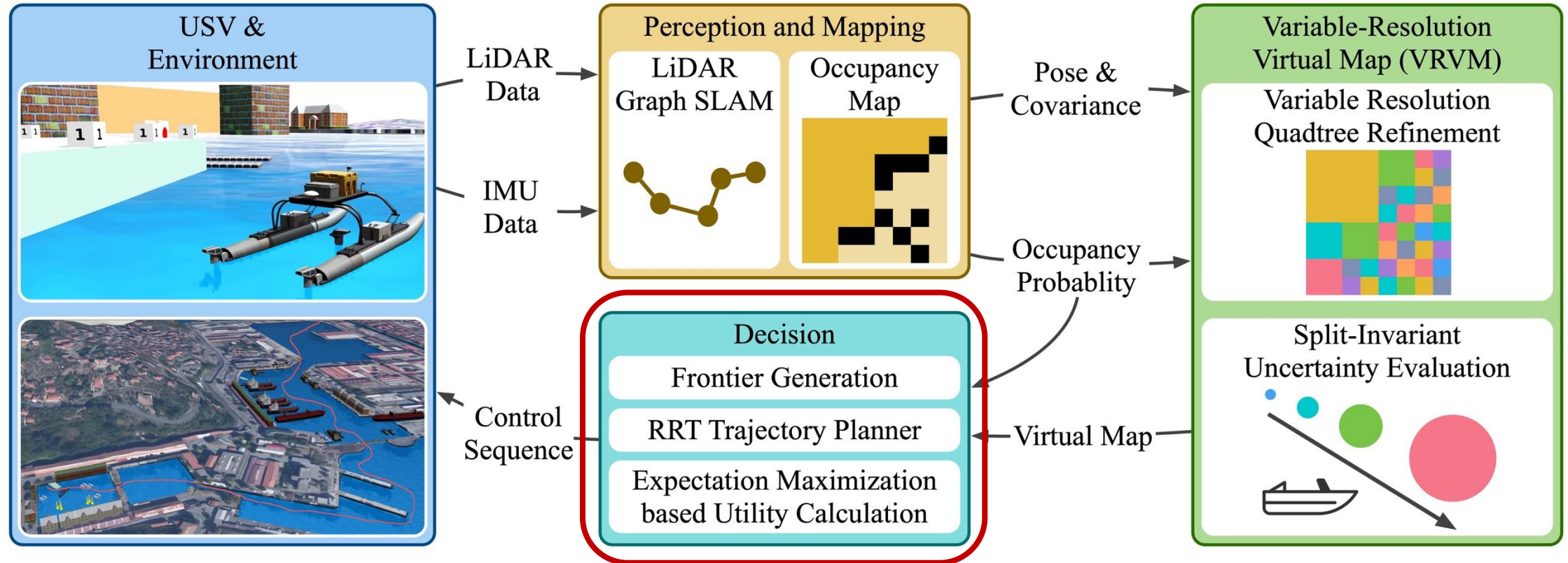
- A quadtree represents several scales.
- Refine selected regions while keeping other regions coarse.

Adapt the representation to the task.



Step 3: Choose where to go next

Generate goals, sample routes, and evaluate their expected utility.



Execute part of the route, collect new data, then plan again.

Generate goals and candidate routes

Explore frontiers

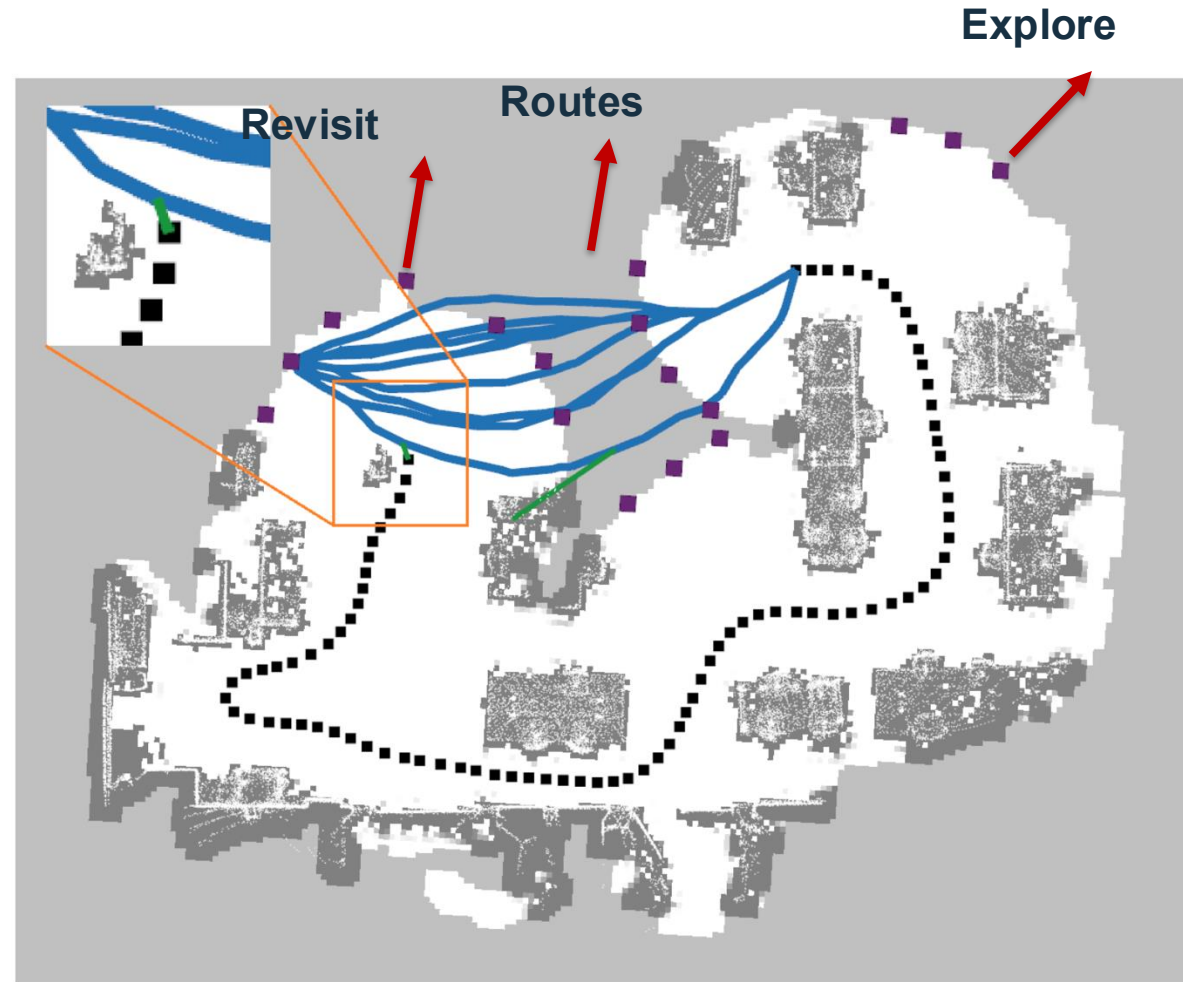
- Reach the boundary of known and unknown space.

Revisit useful areas

- Seek observations that can reduce uncertainty.

Sample routes with RRT

- Evaluate alternatives before moving.



RRT: rapidly exploring random tree

How does the robot choose its next route?

Evaluate each candidate route before committing to it.

$$\text{Route utility} = \text{map term} + \text{trajectory term} + \text{travel term}$$

Map uncertainty

Will the route improve what the robot knows about the environment?

Trajectory uncertainty

Will the robot remain well localized as it moves?

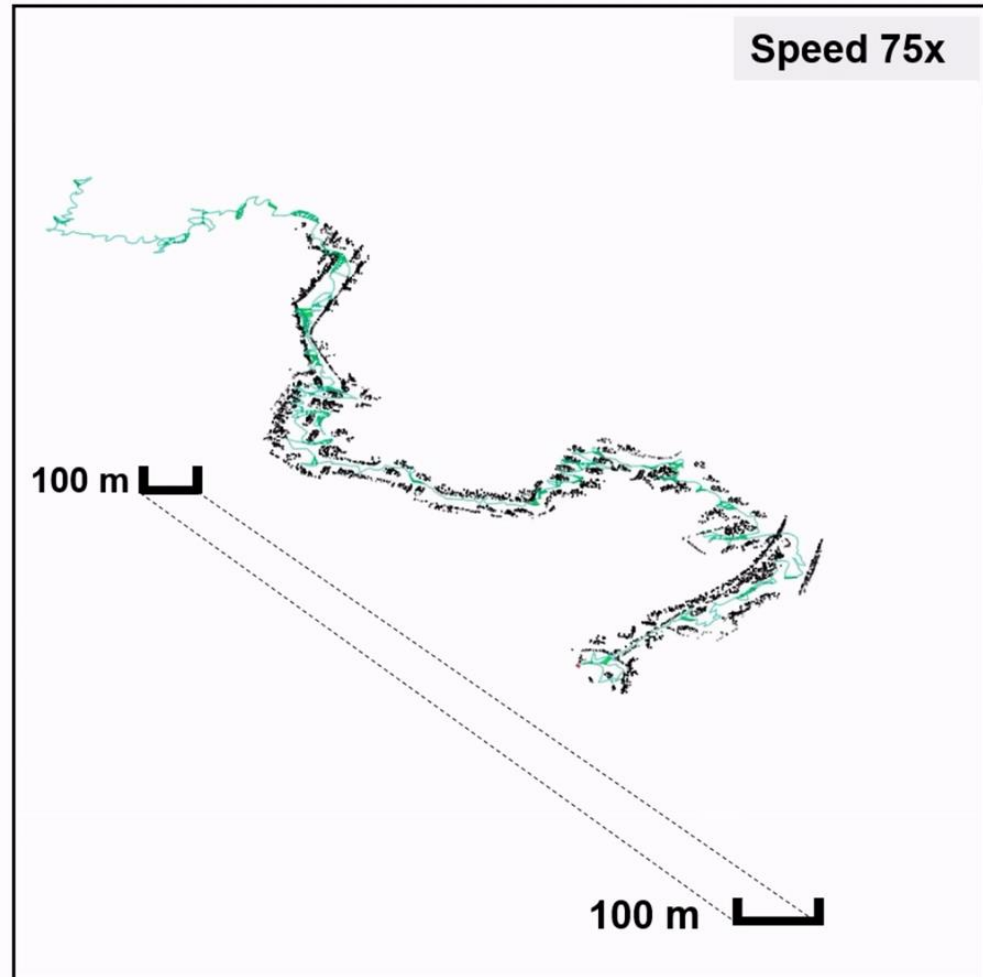
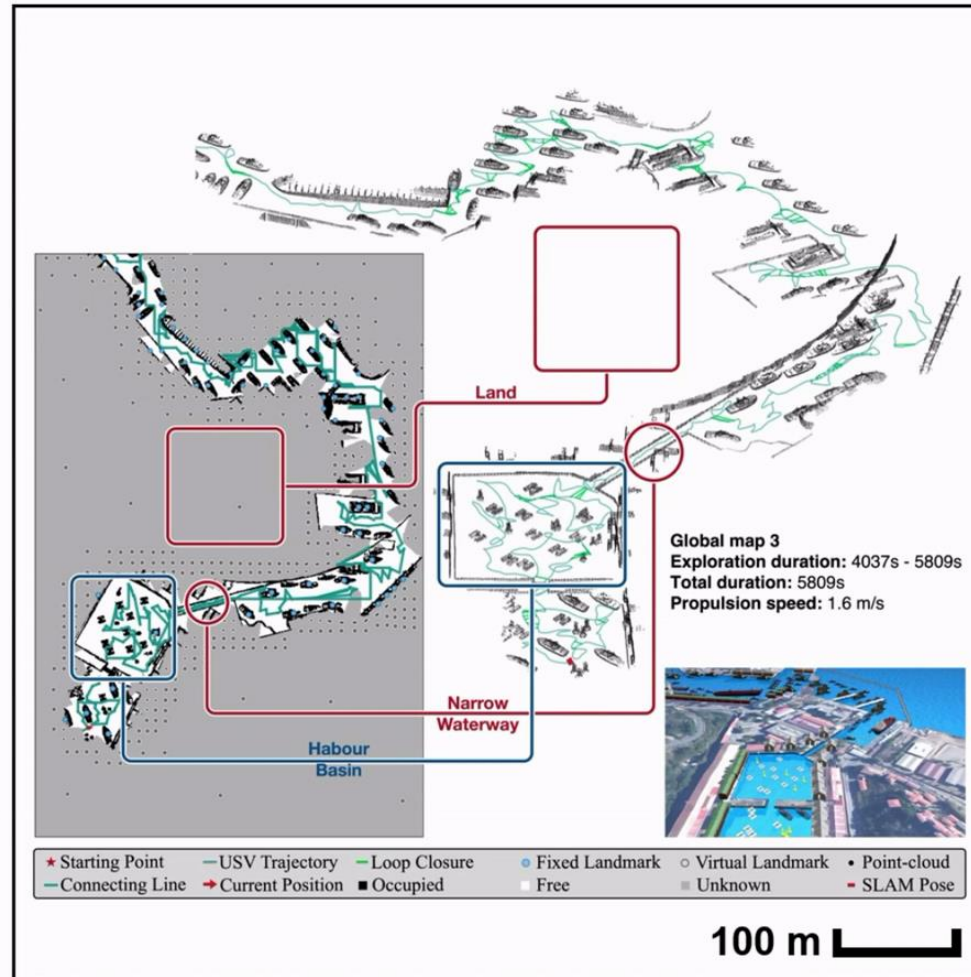
Travel distance

How much motion is needed to obtain that information?

Choose the highest-utility candidate, move, observe, and plan again.

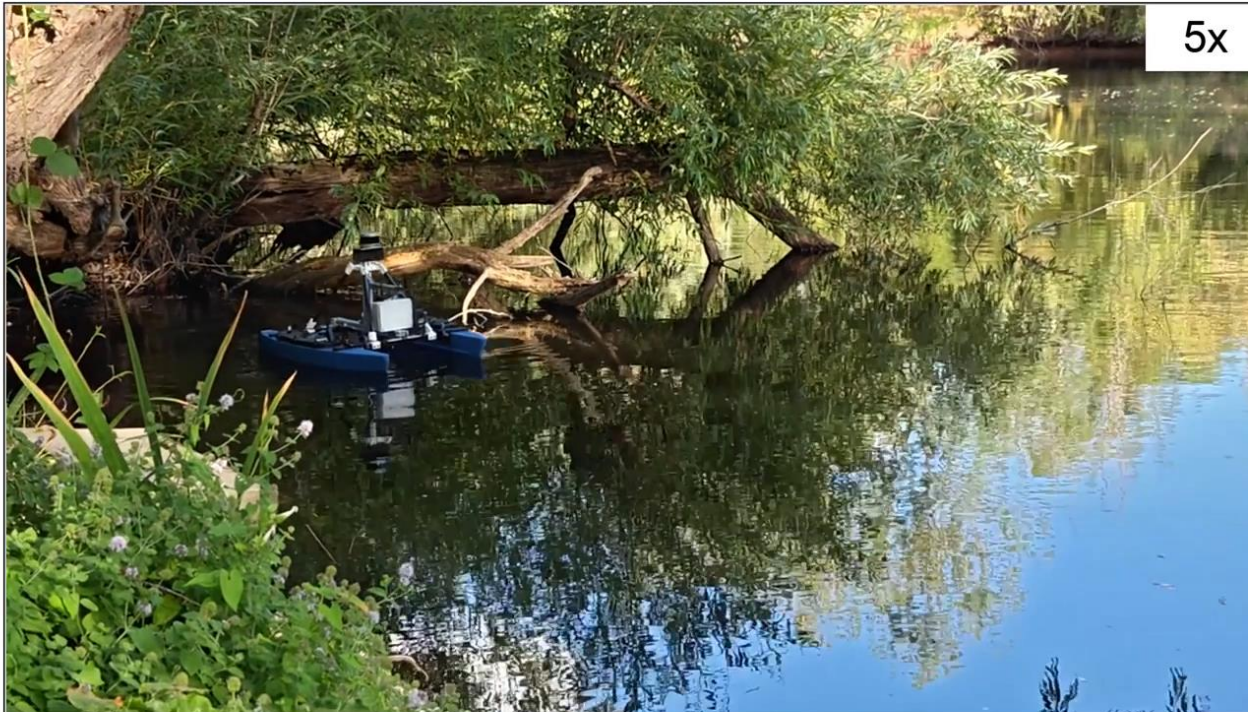
Simulation: autonomous exploration

Exploration in 1000 m × 1000 m Environment



Field operation: onboard planning, manual steering

Field Experiments: Onboard Sensing



VLP-16 LiDAR and MicroStrain IMU

Onboard measurements support online state estimation

Onboard sensors

- 1 LiDAR
Velodyne VLP-16



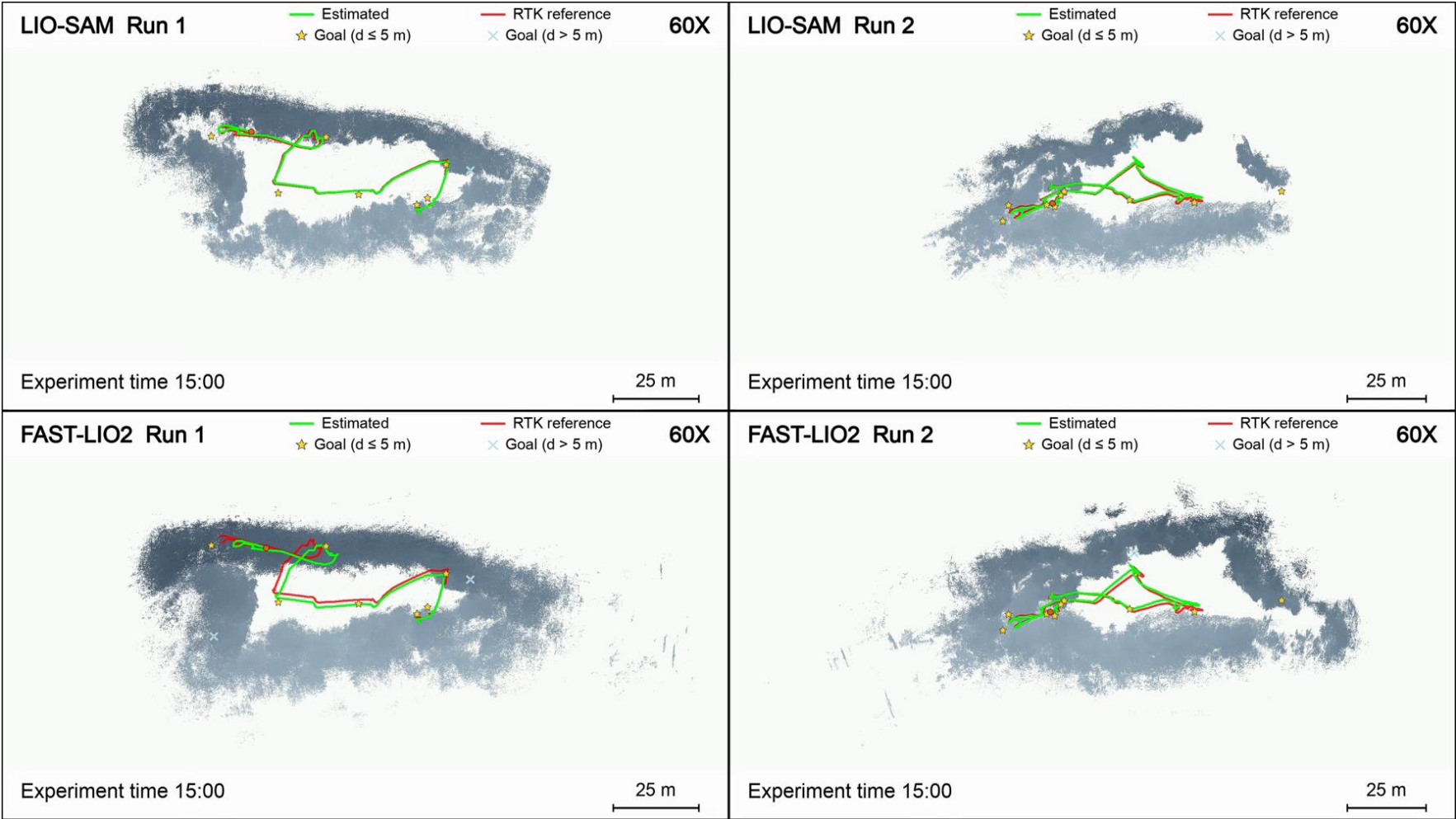
- 2 IMU
MicroStrain
3DM-GX5-AHRS



LiDAR and inertial sensing

Field trials: estimated paths and RTK reference

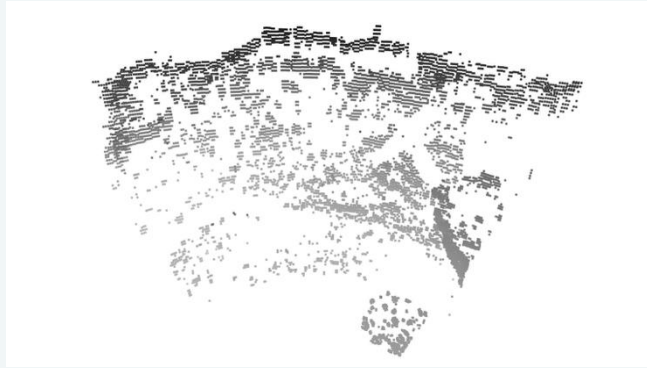
Field Experiments: VRVM-Based Exploration Results



Green: estimated path. Red: RTK reference. · Source playback: 60× · 30 s clip

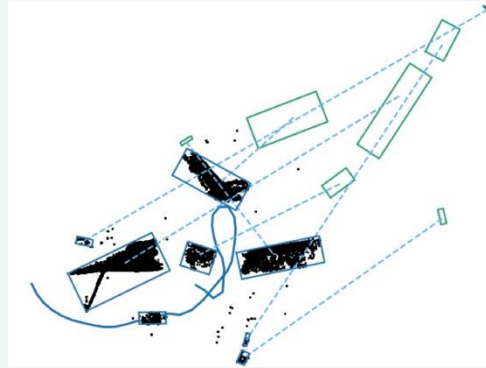
Summary: three takeaways

SLAM connects pose and map



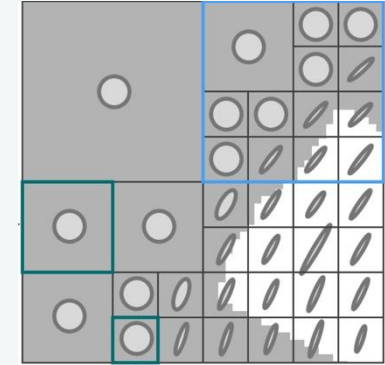
- Combine motion and scene observations.
- Use loop closures to reduce accumulated drift.

Marine SLAM needs reliable matches



- Fuse complementary sensors.
- Match sonar structure and verify inter-robot loop closures.

Exploration should account for uncertainty



- Use virtual maps to predict route uncertainty.
- Balance information, localization, and travel.
- Adapt resolution to limit computation.

Sense → Estimate → Plan → Move → Repeat